Limit shapes for centrally excited walks

Ahmed Bou-Rabee, Tuan-Minh Nguyen, Yuval Peres, and Arvind Singh

Abstract. A centrally excited random walk on $\mathbb{Z}^d$ moves like simple random walk, except that its first step from each site has a drift of fixed size toward the origin. Kozma (2007) conjectured that after n steps the visited set approximates a ball with radius of order $n^{1/(d+1)}$. We prove this in every dimension $d \geq 2$ and show that the rescaled numbers of visits converge uniformly to a cone, which is the potential generated by the drift on a ball. The same argument shows that a drift opposite to a subgradient of a norm produces the ball of that norm. In the plane, the outer and inner radii of the visited set differ by at most $n^{1/6}$ times a power of $\log n$, and the exponent $1/6$ cannot be lowered.

1. Introduction

Let $d \geq 2$ and $0 < \kappa < 1/d$. A **centrally excited random walk** $X_0 = 0, X_1, X_2, \ldots$ on $\mathbb{Z}^d$ moves like simple random walk, except that on its first departure from each site $x \neq 0$ it moves to $x \pm e_i$ with probability $1/(2d) \mp \kappa x_i/(2|x|)$ for $1 \leq i \leq d$, where $e_1, \ldots, e_d$ is the standard basis of $\mathbb{R}^d$. Given the path up to that departure, this step has mean $-\kappa x/|x|$, a drift of size κ toward the origin (Figure 1).

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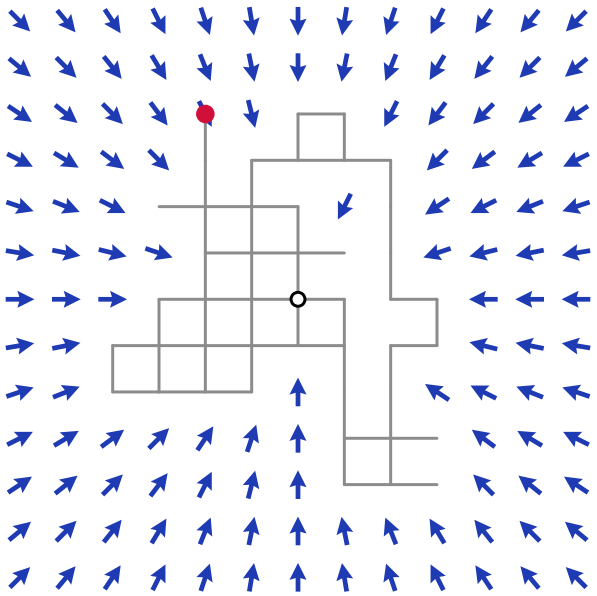


Figure 1. The walk after 120 steps, for $d = 2$ and $\kappa = 0.4$: its path (gray), the origin (open circle) and its position (red dot). At each site $x \neq 0$ that it has not yet left, an arrow shows the mean $-\kappa x/|x|$ of its first step from x.

At an Oberwolfach problem session, Kozma [2007] conjectured that the visited set approximates a ball of radius of order $n^{1/(d+1)}$. We prove this for every $d \geq 2$, identify the radius and the uniform limit of the local times, and bound the fluctuations of both. In the plane the inner and outer radii of the visited set fluctuate on the scale $n^{1/6}$ up to logarithmic factors; in higher dimensions their fluctuations are bounded by powers of $\log n$. More generally, a first-departure drift opposite to a subgradient of a norm produces the ball of that norm.

1.1. **The limit shape.** For an integer $n \geq 0$ let

$$\ell_n(x) \coloneqq \sum_{j=0}^{n-1} \mathbf{1}_{\{X_j = x\}} \qquad \text{and} \qquad A_n \coloneqq \{x : \ell_n(x) > 0\}\,.$$

The **local time** $\ell_n(x)$ is the number of departures from x before time n, and A_n is the **range**; the set of sites visited by time n is A_{n+1}. Write $B(y,\rho)$ for the open Euclidean ball of radius ρ about y and ω_d for the volume of $B(0,1)$, and let

$$r_n \coloneqq \left(\frac{(d+1)n}{2d\kappa\omega_d}\right)^{\frac{1}{d+1}}\,, \tag{1.1}$$

so that the cone $2d\kappa(r_n - |v|)_+$ has integral n. Our first theorem says that the range approximates $B(0,r_n)$ and the local times approximate this cone.

Theorem 1.1 (Limit shape)**.** *Let $d \geq 2$ and $0 < \kappa < \frac{1}{d}$. Then, almost surely, the centrally excited random walk has the following properties.*

(i) Shape*: for every $0 < \eta < 1$ and all sufficiently large n,*

$$\{x \in \mathbb{Z}^d : |x| < (1-\eta)r_n\} \subset A_n \subset \{x \in \mathbb{Z}^d : |x| < (1+\eta)r_n\}\,.$$

(ii) Local times*: as $n \to \infty$,*

$$\frac{1}{r_n} \max_{x \in \mathbb{Z}^d} \left|\ell_n(x) - 2d\kappa(r_n - |x|)_+\right| \longrightarrow 0\,.$$

(iii) Recurrence*: the walk visits every site of $\mathbb{Z}^d$ infinitely often.*

1.2. **Fluctuations.** Replace each site $x \in A_n$ by the unit cell $C_x \coloneqq x + [-\frac{1}{2},\frac{1}{2})^d$, and let $D_n \coloneqq \bigcup_{x \in A_n} C_x$. Define the **inner radius** and the **outer radius** as

$$R_{\text{in}}(n) \coloneqq \inf_{y \notin D_n} |y| \qquad \text{and} \qquad R_{\text{out}}(n) \coloneqq \max_{0 \leq j \leq n} |X_j|\,. \tag{1.2}$$

Theorem 1.2 (Fluctuation bounds)**.** *Let $d \geq 2$ and $0 < \kappa < 1/d$. For every $p > 0$ there exists $C(d,\kappa,p) < \infty$ such that, for every integer $n \geq 2$, the following bounds hold simultaneously with probability at least $1 - Cn^{-p}$.*

(i) Radii*:*

$$\begin{aligned} &|R_{\text{in}}(n) - r_n| \leq C\sqrt{r_n \log n} \quad \textit{and} \quad R_{\text{out}}(n) - r_n \leq C\sqrt{r_n}(\log n)^{5/2} \quad \textit{if } d = 2\,,\\ &|R_{\text{in}}(n) - r_n| \leq C \log n \quad \textit{and} \quad R_{\text{out}}(n) - r_n \leq C(\log n)^{d+1} \quad \textit{if } d \geq 3\,. \end{aligned}$$

(ii) Volume*:*

$$\left|r_n^{-1} D_n \mathbin{\triangle} B(0,1)\right| \leq C \begin{cases} \sqrt{\dfrac{\log n}{r_n}}, & d = 2,\\ \dfrac{\log n}{r_n}, & d \geq 3 \end{cases}\,.$$

(iii) Local times*:*

$$\max_{x\in\mathbb{Z}^d}\big|\ell_n(x) - 2d\kappa(r_n - |x|)_+\big| \le C\begin{cases}\sqrt{r_n}(\log n)^{3/2}, & d=2,\\ \sqrt{r_n\log n}, & d\ge 3\end{cases}.$$

In particular, these bounds hold for all sufficiently large n almost surely.

In the plane, r_n has order $n^{1/3}$, so part (i) bounds the deviations of both radii from r_n by $n^{1/6}$ times a power of $\log n$; in higher dimensions it bounds them by a power of $\log n$. The next theorem shows that the planar exponent $1/6$ cannot be lowered and that the local-time bound in part (iii) has the correct order in $d \ge 3$.

Theorem 1.3 (Sharpness)**.** *Let $d \ge 2$ and $0 < \kappa < 1/d$, and consider the centrally excited random walk.*

(i) Radii in the plane*: let $d = 2$. For every $p > 0$ there exists $c(\kappa, p) > 0$ such that, for all sufficiently large n,*

$$\mathbb{P}\big(r_n - R_{\text{in}}(n) \ge c\sqrt{r_n\log n}\big) \ge n^{-p} \qquad \text{and} \qquad \mathbb{P}\big(R_{\text{out}}(n) - r_n \ge c\sqrt{r_n\log n}\big) \ge n^{-p}. \tag{1.3}$$

(ii) Iterated logarithm for the radii in the plane*: let $d = 2$. Almost surely,*

$$\limsup_{n\to\infty}\frac{r_n - R_{\text{in}}(n)}{\sqrt{r_n\log\log n}} \ge \frac{1}{\sqrt{10\pi\kappa}} \qquad \text{and} \qquad \limsup_{n\to\infty}\frac{R_{\text{out}}(n) - r_n}{\sqrt{r_n\log\log n}} \ge \frac{1}{\sqrt{10\pi\kappa}}. \tag{1.4}$$

(iii) Local times near the origin*: there exists $c(d, \kappa) > 0$ such that, for all sufficiently large n, with probability at least $1 - n^{-c}$,*

$$\max_{\substack{x\in\mathbb{Z}^d\\ |x|\le r_n^{1/2}}}\big|\ell_n(x) - 2d\kappa(r_n - |x|)\big| \ge c\begin{cases}\sqrt{r_n}\log n, & d=2,\\ \sqrt{r_n\log n}, & d\ge 3\end{cases}. \tag{1.5}$$

(iv) Difference of the radii in the plane*: let $d = 2$.*

(a) *For every $p > 0$ there exists $c(\kappa, p) > 0$ such that, for all sufficiently large n,*

$$\mathbb{P}\big(R_{\text{out}}(n) - R_{\text{in}}(n) \ge c\sqrt{r_n\log n}\big) \ge n^{-p}. \tag{1.6}$$

(b) *Almost surely,*

$$\limsup_{n\to\infty}\frac{R_{\text{out}}(n) - R_{\text{in}}(n)}{\sqrt{r_n\log\log n}} \ge \sqrt{\frac{\pi}{3\kappa}}. \tag{1.7}$$

(c) *For every $a \ge 0$,*

$$\limsup_{n\to\infty}\mathbb{P}\big(R_{\text{out}}(n) - R_{\text{in}}(n) \le a\sqrt{r_n}\big) \le 1 - \exp\Big(-\frac{3\kappa a^2}{\pi}\Big). \tag{1.8}$$

Part (i) shows that the bound on $|R_{\text{in}}(n) - r_n|$ in Theorem 1.2 (i) is attained, up to the constant, with probability at least n^{-p}. Part (iv) shows that the planar fluctuations are not only fluctuations of a radius: by (c), the outer and inner radii differ by more than $a\sqrt{r_n}$ with probability at least $e^{-3\kappa a^2/\pi} - o(1)$. Part (iii) matches the local-time bound of Theorem 1.2 (iii) when $d \ge 3$; in the plane we show in Section 9 that, away from the boundary, the local times deviate from the cone by at most $C\sqrt{r_n}\log n$, so part (iii) is sharp there as well.

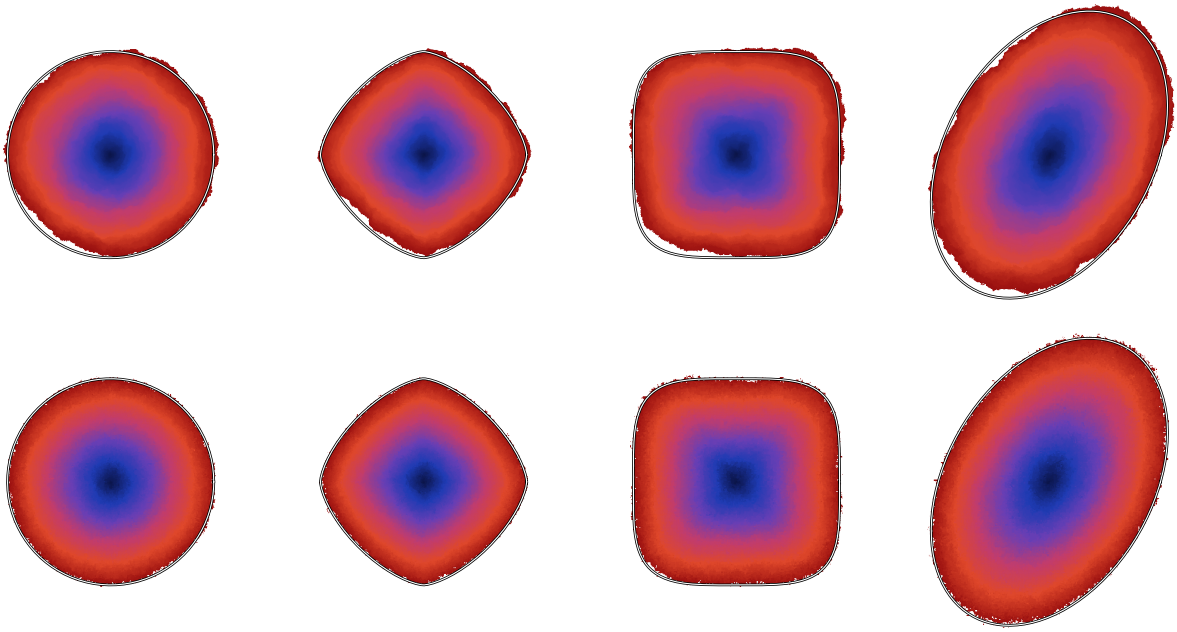


FIGURE 2. Ranges $r_n^{-1}A_n$ for the Euclidean, $\ell^{3/2}$, ℓ^4 and elliptical norms (left to right), at $n = 10^9$. Top: $d = 2$, $\kappa = 0.4$; bottom: $x_3 = 0$ sections for $d = 3$, $\kappa = 0.25$. The unit balls are outlined and the blue-to-red colors show $(t_x/n)^{1/(d+1)}$, with t_x the first-departure time.

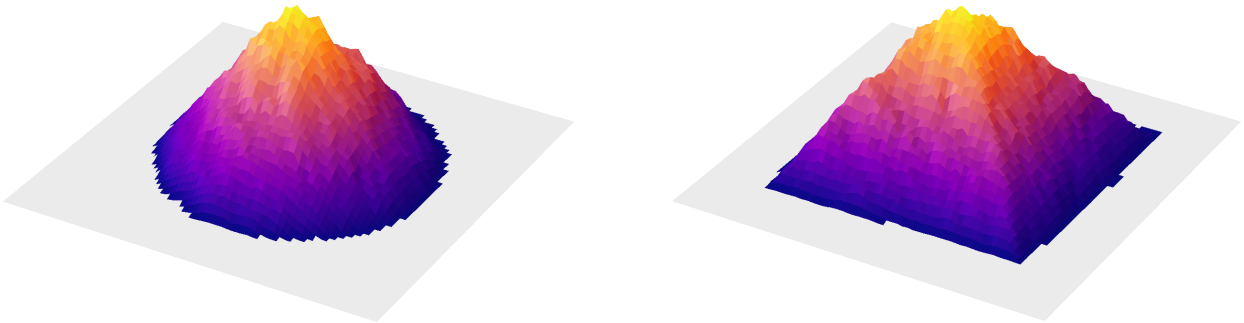


FIGURE 3. Local times $\ell_n(x)/(2d\kappa r_n)$ over x/r_n for the Euclidean norm (left) and ℓ^∞ (right): $d = 2$, $n = 10^{10}$, $\kappa = 0.4$.

1.3. **Limit shapes for other norms.** The limit shape theorem extends to drifts given by other norms. Let Ψ be a norm with unit ball B_Ψ. Replace $x/|x|$ in the first-departure probabilities by a subgradient $\xi(x)$ of Ψ at x, chosen in advance at each site, and assume $\kappa \max_i \Psi(e_i) < 1/d$. Then the range approximates $\{\Psi < r_n\}$ and the local times approximate $2d\kappa(r_n - \Psi)_+$, with $|B_\Psi|$ replacing ω_d in (1.1) (Theorem 2.1).

The theorem includes norms with corners, such as $\|\cdot\|_1$ and $\|\cdot\|_\infty$, with rates uniform in the choice of subgradients. It also extends to the Minkowski functional of every compact convex set containing the origin in its interior (end of Section 5). Figures 2 and 3 show the range and the local times.

1.4. **Strategy and key ideas.** The proof compares the local times with an explicit potential. Dynkin's formula for the Green function with pole y writes $\ell_n(y)$ as a sum over the first departures,

each weighted by the inner product of its drift with the gradient of the Green function, plus a martingale and a smaller error. The continuum version of this sum is a potential: for a bounded measurable set $D \subset \mathbb{R}^d$ and $y \in \mathbb{R}^d$, let

$$U_D(y) := \frac{2\kappa}{\omega_d} \int_D \frac{v}{|v|} \cdot \frac{v-y}{|v-y|^d} \, \mathrm{d}v \tag{1.9}$$

be the potential generated by the vector field $-\kappa v/|v|$ on D. Integration along the rays from y shows that the potential of a ball is a cone: for every $\rho > 0$ and every $y \in \mathbb{R}^d$,

$$U_{B(0,\rho)}(y) = 2d\kappa(\rho - |y|)_+ \,. \tag{1.10}$$

Replacing $v/|v|$ by $\nabla\Psi(v)$ gives the cone $2d\kappa(\rho - \Psi(y))_+$ for a ball of the norm (Lemma 2.2).

So Dynkin's formula compares ℓ_n with U_{D_n}, up to a martingale and a smaller error. Near the boundary of the range, the martingale error bound can be larger than the local times and the potential, so this comparison alone need not locate the boundary. We use two further facts. Local times vanish outside the range, and the walk can reach a distant site only by crossing the intervening region against its first-departure drift. The main comparison is at a contact point: a point y_0 that lies both on the boundary of the largest ball about the origin contained in D_n and in the closure of a cell disjoint from D_n. The local time vanishes on that cell, so the potential at y_0 is small, while every point of D_n outside the ball contributes nonnegatively to it.

1. *Local times and the potential* (Section 3). Freedman's inequality controls the martingales in Dynkin's formula uniformly over sites. Replacing the sum over first departures by U_{D_n} costs $O(\log n)$, also for norms with corners; convexity controls the variation of the subgradients across a cell (Lemma 3.3).
2. *Crossing and coarse bounds* (Section 4). Let τ_R be the time at which the walk leaves the ball of the norm with radius R. Until τ_R, and until one of the martingales in Dynkin's formula exceeds R in absolute value, Dynkin's formula bounds each local time by CR, and hence the elapsed time by CR^{d+1}; Freedman's inequality then shows that the martingales are unlikely to exceed R before τ_R. At time τ_R, the comparison at the contact point and the crossing argument show that the range contains all sites of a ball of radius cR, and that the local times have order R on the concentric ball of half this radius; so $\tau_R \geq cR^{d+1}$, and the coarse bounds at time n follow (Proposition 4.1). The crossing bound comes from the exponential of the distance to a ball of the norm: far from the ball, the drift at the first departure from each site outweighs the second-order terms of all the departures from that site, as long as the local time there is small.
3. *The limit shape* (Section 5). Let b be the radius of the largest ball of the norm about the origin contained in D_n. By convexity, every point v of D_n outside this ball contributes at least $c(\Psi(v) - b)r_n^{-d}$ to the potential at the contact point. Dynkin's formula at the center of the adjacent cell bounds this potential by the martingale there, and Dynkin's formula for the square of the Green function bounds the bracket of that martingale in terms of the local time at its pole, which vanishes (Lemma 5.2). So the part of D_n outside the ball has small potential, which gives the cone for the local times, and the martingale obtained from $|X_n|^2$ gives $b \approx r_n$. The local times outside $\{\Psi < r_n\}$ are then small, and the crossing argument shows that no visited site lies far from $\{\Psi \leq r_n\}$.
4. *The inner radius* (Section 6). For the Euclidean norm, every point v of D_n outside the inner ball contributes at least $c|v - y_0|^{2-d}/r_n$ to the potential at the contact point y_0, a bound that does not vanish near the boundary of the ball. With Lemma 5.2, this bounds the volume of D_n outside the ball and sharpens the inner-radius estimate (Proposition 6.1).
5. *The outer radius* (Section 7). We evaluate the potential at points beyond the outer radius, where the local time vanishes. Points of D_n near the outer sphere contribute negatively to this potential. The curvature of the sphere bounds the positive contributions by an integral of $|v - y|^{2-d}/|v|$ over the part of D_n outside the inner ball, which we bound by integrating the potential along

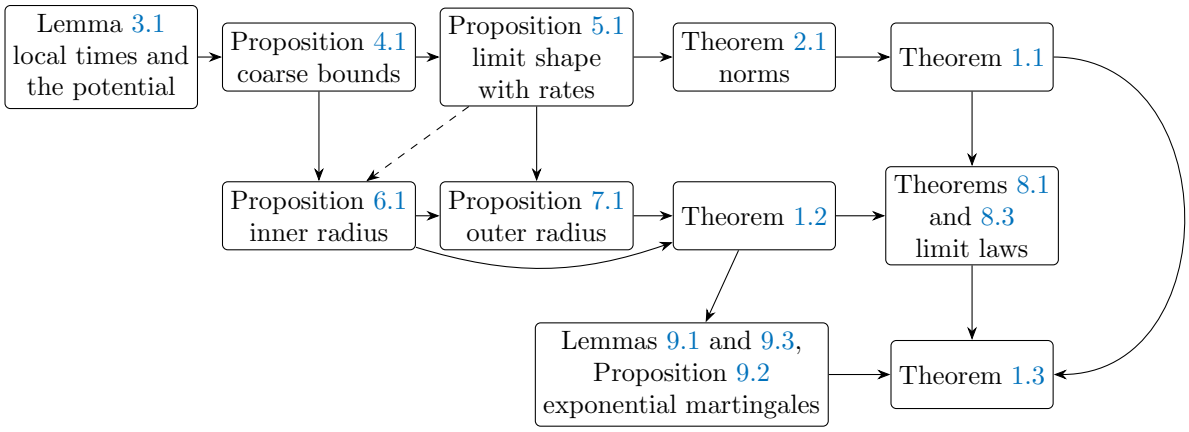


FIGURE 4. Dependencies among the main results. A solid arrow means that a result is used; a dashed arrow means that estimates from Section 5 are used.

rays when $d \geq 3$ and by the volume bound of Section 6 when $d = 2$. So little of D_n lies near the outer sphere. These volume bounds control the local times outside the inner ball, and the crossing argument gives the outer-radius bound (Proposition 7.1). The same estimates and the maximum principle for the potential of this part of D_n give the uniform local-time bound.

The cone determines the limits of the brackets of the martingales in Dynkin's formula and of the martingale obtained from $|X_n|^2$, so the martingale central limit theorem and the law of the iterated logarithm give the limit laws of Section 8. The lower bounds need deviations of about $(\log n)^{1/2}$ standard deviations, which we obtain from exponential martingales: the martingale obtained from $|X_n|^2$ controls the planar radii, and many weakly correlated martingales in Dynkin's formula control the largest deviation of the local times near the origin. The difference of the radii is controlled by the vector martingale $X_n + \kappa \sum_{x \in A_n} x/|x|$: the vectors $x/|x|$ cancel inside the inner disk, so fluctuations of order $\sqrt{n}$ of this martingale force fluctuations of order $\sqrt{r_n}$ of the difference of the radii, and its two-dimensional Gaussian limit gives the bound in Theorem 1.3 (iv)(c). Figure 4 records the dependencies.

The local times play the role of the odometer of the divisible sandpile, which Levine and Peres [2009] compared with an explicit potential. Martingales with a pole at an unvisited site, and separate estimates that rule out thin tentacles, also appear in the bounds of Jerison, Levine, and Sheffield [2012, 2013] for internal diffusion-limited aggregation. Our logarithmic lower bounds use the inner/outer deviation dichotomy of Asselah and Gaudillière [2014, Proposition 1.2].

1.5. **Related work.** Kozma [2013] reported recurrence of the walk excited to the center, citing unpublished work from 2006. Dembo, Huang, and Sidoravicius [2014a] described the manuscript of Kozma [2006] for a related walk in $d \geq 3$, with drift $-\kappa x/\|x\|_1$ at the first visit to a site having an unvisited neighbor; Dembo, Huang, and Sidoravicius [2014b] cited it for recurrence in every dimension. We take the name centrally excited random walk from the title of that manuscript. The drift of its walk has a Euclidean length that depends on the direction, whereas our drift has length κ. Benjamini, Duminil-Copin, Kozma, and Lucas [2020] also recorded the predicted radius $n^{1/(d+1)}$.

For the excited random walk with first-visit drift in a fixed direction, Benjamini and Wilson [2003] proved transience in $d \geq 2$ and positive lower speed in $d \geq 4$. Kozma [2003, 2005] obtained positive lower speed in $d = 3$ and $d = 2$, and Bérard and Ramírez [2007] proved laws of large numbers and central limit theorems. Menshikov, Popov, Ramírez, and Vachkovskaia [2012] extended positive-speed results to uniformly elliptic walks with bounded jumps and uniformly positive first-visit drift in a fixed direction and zero drift at later visits. Zerner [2005, 2006] gave recurrence/transience criteria on $\mathbb{Z}$ and strips for walks with multiple cookies. Amir, Benjamini, and Kozma [2008] proved

recurrence for a walk in the upper half of $\mathbb{Z}^3$ that steps down at new sites. See Kosygina and Zerner [2013] for a survey.

The radius $n^{1/(d+1)}$ and range size $n^{d/(d+1)}$ are also predicted for once-reinforced walks with strong reinforcement [Beffara, 2011; Elboim and Kozma, 2026], and supported by simulations [Foster, Grassberger, and Paczuski, 2009]; related predictions for self-attracting walks appear in Ordemann, Berkolaiko, Havlin, and Bunde [2000]; Ordemann, Tomer, Berkolaiko, Havlin, and Bunde [2001]. For once-reinforced random walk on $\mathbb{Z}^d$ with $d \geq 2$, Bou-Rabee and Peres [2026c] proved the lower bound $cn^{d/(d+1)}$ on the expected range for every reinforcement parameter. The matching upper bound and limit shape remain conjectural. See Pemantle [2007] for reinforced processes.

Dembo, Groisman, Huang, and Sidoravicius [2021] proved an averaging principle and shape theorem for growth models motivated by origin-excited and once-reinforced walks. Their simulations with inward first steps along a coordinate of largest absolute value or along all coordinates produced squares and diamonds. Our norm theorem proves these shapes for first-departure drifts opposite to subgradients of $\|\cdot\|_\infty$ and $\|\cdot\|_1$. For related continuum growth models, Dembo and Yang [2025a,b] obtained limits described by geometric flow and regularized Kardar–Parisi–Zhang equations.

Internal diffusion-limited aggregation [Lawler, Bramson, and Griffeath, 1992] and rotor-router aggregation [Levine and Peres, 2009] have Euclidean ball limit shapes. The former has radius fluctuations of order at most $\log r$ in the plane and $(\log r)^{1/2}$ in higher dimensions [Jerison et al., 2012, 2013; Asselah and Gaudillière, 2013a,b], and its rescaled fluctuations approach a variant of the Gaussian free field [Jerison, Levine, and Sheffield, 2014]. For the centrally excited walk, Theorem 1.3 gives much larger planar fluctuations of the radii and of their difference; the fixed-site covariance in Theorem 8.3 is again expressed through the Green function. See Levine and Peres [2017] for aggregation models.

For Eulerian walkers [Priezzhev, Dhar, Dhar, and Krishnamurthy, 1996], simulations suggested a circular planar range of radius of order $n^{1/3}$ after n steps [Kapri and Dhar, 2009]. Florescu, Levine, and Peres [2016] proved that every simple rotor walk on $\mathbb{Z}^d$ visits at least $cn^{d/(d+1)}$ sites in n steps. For clockwise planar rotors with independent uniform initial states, Bou-Rabee and Peres [2026a] proved convergence of the rescaled range to a deterministic compact convex set; whether it is a disk remains open.

1.6. **Notation and conventions.**

- The Euclidean norm is $|x|$, and $\|x\|_p$ is the ℓ^p norm for $1 \leq p \leq \infty$. The cardinality of a set $A \subset \mathbb{Z}^d$ is $|A|$, and the Lebesgue measure of a set $D \subset \mathbb{R}^d$ is $|D|$. The symmetric difference is $\triangle$, and $t_+ := \max\{t, 0\}$.
- Let $u_x := x/|x|$ for $x \in \mathbb{R}^d \setminus \{0\}$ and $u_0 := 0$, and let $\xi(0) := 0$.
- The letters $c > 0$ and $C > 0$ denote constants depending only on d, κ and the exponent p of the failure probability under discussion, and in Sections 2–5 also on the norm Ψ, but not on the choice of the subgradients $\xi(x)$; they may change from line to line. Every further dependence is stated, and a subscript on O, as in O_y, means that the implied constant also depends on y. We write $a_n \asymp b_n$ if $c \leq a_n/b_n \leq C$.
- The σ-field generated by $X_0, \dots, X_j$ is $\mathcal{F}_j$. For martingales Z_1 and Z_2, the brackets $\langle Z_1 \rangle_n$ and $\langle Z_1, Z_2 \rangle_n$ are the predictable quadratic variation and covariation through time n.
- The indicator that the step at time j is a first departure is $I_j := \mathbf{1}_{\{X_j \notin A_j\}}$, so that $\mathbb{E}(X_{j+1} - X_j \mid \mathcal{F}_j) = -\kappa I_j \xi(X_j)$, where $\xi(x) = u_x$ for the Euclidean norm.
- The function $\widetilde{\ell}_n$ on $\mathbb{R}^d$ equals $\ell_n(x)$ on C_x for every $x \in \mathbb{Z}^d$; it vanishes off D_n.
- All estimates concern integers $n \geq 2$, so that $\log n > 0$. In a statement that holds with probability at least $1 - Cn^{-p}$ for all sufficiently large n, enlarging C makes it hold for every $n \geq 2$.

1.7. **Lean formalization.** Every labeled theorem, lemma, proposition and corollary of this paper has been formalized in Lean 4. The formalization builds on Mathlib. No proof uses `sorry` or an axiom beyond the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. The development is at https://github.com/nitromannitol/Centrally-Excited-Random-Walk.

Use of artificial intelligence. The authors developed the main ideas of this work and directed its mathematical development. Throughout the project, the authors used large language models as research assistants: to explore specific questions arising in the proofs, to check the arguments and the cited results, and to audit drafts of the paper. The exposition, the simulations and the figures were produced with AI agents, primarily Claude (Anthropic), under the close supervision of the authors. The authors take responsibility for the mathematical content and presentation of the paper.

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Earlier versions and related models. This paper develops the results of Bou-Rabee and Peres [2026b]. The proof of the coarse bounds in Section 4 uses the stopping argument of Singh [2026], who allowed arbitrary first-departure laws with mean $-\kappa x/|x|$ at every nonzero site and a symmetric first-departure law at the origin. Nguyen [2026] considered first-departure laws that favor the neighbors decreasing the ℓ^1 distance to the origin, and proved convergence in probability, in Hausdorff distance, of the rescaled visited set to an explicit ℓ^1 ball in every dimension $d \geq 2$.

2. Drifts given by a norm

In this section we give the precise setting for a drift opposite to a subgradient of a norm and state the limit shape theorem in this setting, Theorem 2.1, of which Theorem 1.1 is the Euclidean case. We also prove the identity for the potential of a ball of the norm on which its proof rests (Lemma 2.2). Sections 3–5 work in this setting and prove Theorem 2.1 at the end of Section 5.

Let Ψ be a norm on $\mathbb{R}^d$ and let B_Ψ be its unit ball. A **subgradient** of Ψ at $x \in \mathbb{R}^d$ is a vector $\xi \in \mathbb{R}^d$ such that $\Psi(y) \geq \Psi(x) + \xi \cdot (y - x)$ for every $y \in \mathbb{R}^d$, and $\partial\Psi(x)$ is the set of subgradients at x; if Ψ is differentiable at x, then $\partial\Psi(x) = \{\nabla\Psi(x)\}$. At each site $x \neq 0$ fix $\xi(x) \in \partial\Psi(x)$, with coordinates $\xi_i(x)$. The **centrally excited random walk with norm** Ψ moves like simple random walk, except that on its first departure from each site $x \neq 0$ it moves to $x \pm e_i$ with probability

$$\frac{1}{2d} \mp \frac{\kappa}{2}\xi_i(x) \qquad \text{for } 1 \leq i \leq d\,. \tag{2.1}$$

We assume that

$$\kappa \max_{1\leq i\leq d} \Psi(e_i) < \frac{1}{d}\,. \tag{2.2}$$

Every $\xi \in \partial\Psi(x)$ satisfies Euler's relation $\xi \cdot x = \Psi(x)$, so the definition of a subgradient gives

$$\xi \cdot y \leq \Psi(y) \qquad \text{for every } y \in \mathbb{R}^d\,, \tag{2.3}$$

with equality at $y = x$. Hence $|\xi_i(x)| \leq \Psi(e_i)$, and (2.2) makes the probabilities (2.1) positive. The step at the first departure from x has conditional mean $-\kappa\xi(x)$, whose inner product with x is $-\kappa\Psi(x) < 0$. The Euclidean norm, with $\xi(x) = x/|x|$, gives the walk of Section 1, and then (2.2) reads $\kappa < 1/d$. Let

$$r_n := \left(\frac{(d+1)n}{2d\kappa|B_\Psi|}\right)^{1/(d+1)}\,, \tag{2.4}$$

which is (1.1) when Ψ is the Euclidean norm; the cone $2d\kappa(r_n - \Psi(v))_+$ has integral n over $\mathbb{R}^d$.

In Sections 2–5, Ψ is a norm, $\kappa > 0$ satisfies (2.2), r_n is given by (2.4), and U_D is the potential (2.5) defined below; at a point $v \in \mathbb{R}^d$, $\nabla\Psi(v)$ is the gradient, which exists for almost every v. Let

$\Lambda_\Psi := \max_{|u|=1} \Psi(u)$ and $c_\Psi := \min_{|u|=1} \Psi(u)$; by (2.3), every subgradient ξ of Ψ satisfies $|\xi| \le \Lambda_\Psi$. In Sections 6–9, Ψ is the Euclidean norm, so that $\xi(x) = u_x$, r_n is given by (1.1), and U_D is the potential (1.9).

Theorem 2.1 (Limit shape for a norm)**.** *Let $d \ge 2$, let Ψ be a norm on $\mathbb{R}^d$, let $\xi(x) \in \partial\Psi(x)$ be chosen at the sites $x \in \mathbb{Z}^d \setminus \{0\}$, let $\kappa > 0$ satisfy (2.2), and let r_n be as in (2.4). Then, almost surely, the centrally excited random walk with norm Ψ has the following properties.*

(i) Shape*: for every $0 < \eta < 1$ and all sufficiently large n,*

$$\{x \in \mathbb{Z}^d : \Psi(x) < (1-\eta)r_n\} \subset A_n \subset \{x \in \mathbb{Z}^d : \Psi(x) < (1+\eta)r_n\}\,.$$

(ii) Local times*: as $n \to \infty$,*

$$\frac{1}{r_n} \max_{x \in \mathbb{Z}^d} \bigl|\ell_n(x) - 2d\kappa(r_n - \Psi(x))_+\bigr| \longrightarrow 0\,.$$

(iii) Recurrence*: the walk visits every site of $\mathbb{Z}^d$ infinitely often.*

Proposition 5.1 gives rates: for every $p > 0$ there is a constant $C(d, \kappa, \Psi, p)$, not depending on the choice of ξ, such that with probability at least $1 - Cn^{-p}$ the maximum in part (ii) is at most $Cr_n^{5/6}(\log n)^{1/6}$ if $d = 2$ and $Cr_n^{d/(d+1)}(\log n)^{1/(d+1)}$ if $d \ge 3$, and the inclusions of part (i) hold with ηr_n replaced by $\log n$ times this bound on the maximum. Part (i) gives $|A_n|/r_n^d \to |B_\Psi|$, and for the Euclidean norm Theorem 1.2 gives smaller error bounds.

2.1. **The potential of a norm ball.** For a norm Ψ, the potential (1.9) is defined with the gradient of Ψ in place of $v/|v|$: for a bounded measurable set $D \subset \mathbb{R}^d$ and $y \in \mathbb{R}^d$, let

$$U_D(y) := \frac{2\kappa}{\omega_d} \int_D \nabla\Psi(v) \cdot \frac{v-y}{|v-y|^d}\,\mathrm{d}v \tag{2.5}$$

be the potential generated by the vector field $-\kappa\nabla\Psi$ on D.

For every Lipschitz function f with compact support and every $y \in \mathbb{R}^d$,

$$\int_{\mathbb{R}^d} \nabla f(v) \cdot \frac{v-y}{|v-y|^d}\,\mathrm{d}v = -d\omega_d f(y)\,, \tag{2.6}$$

because in polar coordinates $v = y + tu$ about y the integrand times $\mathrm{d}v$ is $\partial_t f(y+tu)\,\mathrm{d}t\,\mathrm{d}S(u)$, with S the surface measure on the unit sphere, for almost every u by Rademacher's theorem, and S has total mass $d\omega_d$.

Lemma 2.2 (The potential of a norm ball)**.** *Let Ψ be a norm, let $\rho > 0$, and let $y \in \mathbb{R}^d$. Then*

$$U_{\{\Psi<\rho\}}(y) = 2d\kappa(\rho - \Psi(y))_+\,. \tag{2.7}$$

Proof. Apply (2.6) to $f = (\rho - \Psi)_+$, whose gradient is $-\mathbf{1}_{\{\Psi<\rho\}}\nabla\Psi$ almost everywhere. □

For the Euclidean norm, Lemma 2.2 is the identity (1.10).

3. Local times and the potential

In this section we prove Lemma 3.1, which compares the local times with the potential U_{D_n}. Dynkin's formula expresses each local time as U_{D_n} minus a martingale, up to an error term. We bound the bracket of this martingale by a weighted sum of the local times, and use convexity to bound the error from replacing the subgradients $\xi(x)$ chosen at the sites by the gradient $\nabla\Psi$ on the cells (Lemma 3.3).

Lemma 3.1. *Let $p > 0$. There exists $C(d, \kappa, \Psi, p) < \infty$ such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$, every $y \in \mathbb{R}^d$ with $|y| \leq 2n$ satisfies*

$$|\widetilde{\ell}_n(y) - U_{D_n}(y)| \leq C \log n + C \begin{cases} \left(\max_x \ell_n(x)\right)^{1/2} \log n, & d = 2, \\ \left(\max_x \ell_n(x) \log n\right)^{1/2}, & d \geq 3 \end{cases}. \tag{3.1}$$

Once $\max_x \ell_n(x) \leq Cr_n$, as on the event of Proposition 4.1, the right side of (3.1) is at most $C\sqrt{r_n} \log n = o(r_n)$.

3.1. **Dynkin's formula for the local times.** Let $\Delta f(x) := (2d)^{-1} \sum_{|e|=1} (f(x+e) - f(x))$, summed over the unit vectors $e \in \mathbb{Z}^d$, be the discrete Laplacian. For $d = 2$, let g be the potential kernel of simple random walk, normalized by $g(0) = 0$. For $d \geq 3$, let $g := -G$, where the Green function $G(x)$ is the expected number of visits to x by simple random walk from the origin. In both cases $\Delta g = \mathbf{1}_{\{0\}}$; for $d = 2$ see Lawler and Limic [2010, Proposition 4.4.2]. By Lawler and Limic [2010, Theorems 4.3.1 and 4.4.4], as $|x| \to \infty$,

$$g(x) = \begin{cases} \dfrac{2}{\pi} \log |x| + c_g + O(|x|^{-2}), & d = 2, \\ -\dfrac{2}{(d-2)\omega_d} |x|^{2-d} + O(|x|^{-d}), & d \geq 3 \end{cases}, \tag{3.2}$$

where c_g is a constant. Let $\bar{\nabla} g(x)$ be the vector with coordinates $(g(x + e_i) - g(x - e_i))/2$ for $1 \leq i \leq d$. Then

$$\bar{\nabla} g(x) = \frac{2}{\omega_d} \frac{x}{|x|^d} + O(|x|^{-d}) \quad \text{and} \quad |g(x+e) - g(x)| \leq C(1 + |x|)^{1-d}, \tag{3.3}$$

the first for $x \neq 0$, the second for every $x \in \mathbb{Z}^d$ and every unit vector $e \in \mathbb{Z}^d$; see Lawler and Limic [2010, Corollaries 4.3.3 and 4.4.5].

For $f \colon \mathbb{Z}^d \to \mathbb{R}$ with bounded nearest-neighbor increments, let

$$\mathcal{M}_t^f := f(X_t) - f(X_0) - \sum_{j<t} \mathbb{E}\big(f(X_{j+1}) - f(X_j) \,\big|\, \mathcal{F}_j\big)$$

be the martingale in Dynkin's formula for f. By (2.1), the conditional expectation in the sum is $\Delta f(X_j) - \kappa I_j \xi(X_j) \cdot \bar{\nabla} f(X_j)$, with $\bar{\nabla} f$ defined like $\bar{\nabla} g$. For $y \in \mathbb{Z}^d$ write $g_y := g(\cdot - y)$ and $\mathcal{M}^y := \mathcal{M}^{g_y}$. Since $\Delta g_y = \mathbf{1}_{\{y\}}$ and each site of A_n has its first departure before time n,

$$\ell_n(y) = g(X_n - y) - g(-y) + \kappa \sum_{x \in A_n} \xi(x) \cdot \bar{\nabla} g(x - y) - \mathcal{M}_n^y. \tag{3.4}$$

The difference $g(X_n - y) - g(-y)$ is $O(\log n)$ for $d = 2$ and $O(1)$ for $d \geq 3$, uniformly over $y \in \mathbb{Z}^d$ with $|y| \leq 3n$. By (3.3), the increments of $\mathcal{M}^y$ are bounded, and a departure from x adds at most $C(1 + |x - y|)^{2-2d}$ to the bracket of $\mathcal{M}^y$; so, for every integer $t \geq 0$,

$$\langle \mathcal{M}^y \rangle_t \leq C \sum_{j<t} (1 + |X_j - y|)^{2-2d}. \tag{3.5}$$

Lemma 3.2. *Let $p > 0$. There exists $C(p) < \infty$ such that the following holds for every integer $n \geq 2$ and every real martingale Z whose increments have absolute value at most 1: with probability at least $1 - Cn^{-p}$,*

$$|Z_n - Z_0| \leq C\big(\sqrt{\langle Z \rangle_n \log n} + \log n\big). \tag{3.6}$$

Proof. For $0 \leq i \leq \log_2 n + 1$, Freedman [1975, Theorem 1.6] bounds the probability that $\langle Z \rangle_n \leq 2^i$ and $|Z_n - Z_0| > C(p)(\sqrt{2^i \log n} + \log n)$ by $2n^{-p-1}$. Since $\langle Z \rangle_n \leq n$, a union bound over these $O(\log n)$ values of i proves (3.6). ∎

We apply Lemma 3.2 to constant multiples of possibly stopped martingales, with union bounds over polynomially many deterministic sites; the resulting estimates then hold also at sites that depend on the path. In particular, for every $p > 0$, with probability at least $1 - Cn^{-p}$, simultaneously for all $y \in \mathbb{Z}^d$ with $|y| \le 3n$,

$$|\mathcal{M}_n^y| \le C\big(\sqrt{\langle \mathcal{M}^y \rangle_n \log n} + \log n\big)\,. \tag{3.7}$$

Let μ be the distributional Laplacian of Ψ. Since Ψ is convex, μ is a positive Radon measure, and $\int \phi \,\mathrm{d}\mu = -\int \nabla\Psi \cdot \nabla\phi$ for every Lipschitz function ϕ with compact support. Taking $\phi(v) \coloneqq \min\{1, (2 - |v-y|/R)_+\}$ and using $|\nabla\Psi| \le \Lambda_\Psi$ gives, for $y \in \mathbb{R}^d$ and $R > 0$,

$$\mu(B(y,R)) \le 2^d \omega_d \Lambda_\Psi R^{d-1}\,. \tag{3.8}$$

Lemma 3.3. *There exists $C(d) < \infty$ such that, for every $x \in \mathbb{Z}^d$,*

$$\int_{C_x} |\nabla\Psi(v) - \xi(x)| \,\mathrm{d}v \le C\mu\big(B(x, 6\sqrt{d})\big)\,.$$

Proof. The function $h(v) \coloneqq \Psi(v) - \Psi(x) - \xi(x)\cdot(v-x)$ is convex and nonnegative (for $x = 0$ because $\xi(0) = 0$), $h(x) = 0$, and $\nabla h = \nabla\Psi - \xi(x)$ almost everywhere. By convexity along coordinate lines, $|\partial_i h(v)| \le h(v+e_i) + h(v-e_i)$, and the tangent plane at v lies below $h(x) = 0$, so $h(v) \le \nabla h(v)\cdot(v-x)$. Let $A \coloneqq 2\sqrt{d}$ and $\phi(v) \coloneqq (A^2 - |v-x|^2)_+/2$, so that $\nabla\phi(v) = x - v$ on $B(x,A)$ and $\int \nabla h \cdot \nabla\phi = \int \nabla\Psi\cdot\nabla\phi$. Hence

$$\int_{C_x} |\nabla h(v)| \,\mathrm{d}v \le \int_{B(x,A)} h(v)\,\mathrm{d}v \le \int_{B(x,A)} \nabla h(v)\cdot(v-x)\,\mathrm{d}v = \int \phi \,\mathrm{d}\mu \le \frac{A^2}{2}\,\mu\big(B(x,A)\big)\,,$$

which proves the lemma. □

Proof of Lemma 3.1. Let $y \in \mathbb{Z}^d$ with $|y| \le 3n$. We first show the deterministic identity

$$\ell_n(y) = U_{D_n}(y) - \mathcal{M}_n^y + O(\log n)\,. \tag{3.9}$$

In (3.4), the summand at $x = y$ vanishes. For $x \ne y$, replacing $\bar\nabla g(x-y)$ by $(2/\omega_d)(x-y)|x-y|^{-d}$, and then by its average over C_x, costs $O\big(\sum_{|x|\le n}(1+|x-y|)^{-d}\big) = O(\log n)$ by (3.3). Replacing $\xi(x)$ by $\nabla\Psi$ on C_x costs $O(1)$ for the x with $|x-y| \le 8\sqrt{d}$ and, by Lemma 3.3 and the bounded overlap of the balls $B(x, 6\sqrt{d})$, at most $C\int_{\{1\le|z-y|\le 8n\}} |z-y|^{1-d}\,\mathrm{d}\mu(z)$ for the others; by (3.8) each of the $O(\log n)$ dyadic annuli about y contributes $O(1)$. This proves (3.9). Splitting the integrals defining $U_{D_n}(y)$ and $U_{D_n}(z)$ at distance $2\sqrt{d}$ from y gives similarly, for $y, z \in \mathbb{R}^d$ with $|y-z| \le \sqrt{d}$ and $|y|, |z| \le 3n$,

$$|U_{D_n}(y) - U_{D_n}(z)| \le C\log n\,. \tag{3.10}$$

Every site visited before time n lies within distance $4n$ of y, and the weights $(1+|x-y|)^{2-2d}$ over $|x-y| \le 4n$ sum to $O(\log n)$ for $d = 2$ and $O(1)$ for $d \ge 3$. So (3.5) bounds $\langle \mathcal{M}^y\rangle_n$ by $C\max_x \ell_n(x)$ times this sum, and on the event of (3.7), $|\mathcal{M}_n^y|$ is at most the right side of (3.1). For $y \in \mathbb{R}^d$ with $|y| \le 2n$, apply this and (3.9) at the site z with $y \in C_z$, and use (3.10). □

3.2. **Bounds on the potential.** The potential of a set is bounded by its volume, and by the width of a shell of Ψ that contains it. Let $D \subset \mathbb{R}^d$ be a bounded measurable set. For every $y \in \mathbb{R}^d$, the integral of $|v-y|^{1-d}$ over D is at most its integral over the ball of volume $|D|$ centered at y, which is $C|D|^{1/d}$; since $|\nabla\Psi| \le \Lambda_\Psi$, there exists $C(d,\Psi) < \infty$ such that

$$\sup_{y\in\mathbb{R}^d} |U_D(y)| \le C\kappa|D|^{1/d}\,. \tag{3.11}$$

The function U_D is continuous, because it is the convolution of the bounded, compactly supported vector field $\mathbf{1}_D\nabla\Psi$ with a kernel that is the sum of an integrable function and a bounded one.

Convexity of Ψ along lines bounds the potential of a set that lies between two balls of Ψ about the origin.

Lemma 3.4. *Let $b \geq 0$ and $a > 0$, and let D be a measurable subset of $\{v \in \mathbb{R}^d : b \leq \Psi(v) \leq b + a\}$. Then*

$$\sup_{y \in \mathbb{R}^d} |U_D(y)| \leq 2d\kappa a\,.$$

Proof. In polar coordinates $v = y + tu$ about y, as for (2.6), $U_D(y)$ is $2\kappa/\omega_d$ times the integral over unit vectors u of $\int_0^\infty \mathbf{1}_D(y+tu)h_u'(t)\,\mathrm{d}t$, where $h_u(t) \coloneqq \Psi(y+tu)$. The convex function h_u is monotone on each of two intervals, on each of which the substitution $s = h_u(t)$ bounds the integral of $\mathbf{1}_{[b,b+a]}(h_u(t))|h_u'(t)|$ by a. So the inner integral lies in $[-a, a]$, and the unit sphere has area $d\omega_d$. □

4. Coarse bounds on the range and the local times

We prove that, with high probability, $|A_n|$ has order r_n^d, and $R_{\mathrm{out}}(n) = \max_{j \leq n} |X_j|$ and the largest local time have order r_n (Proposition 4.1). Fix $R \geq 2$. Until the walk reaches $\{\Psi \geq R\}$, and until one of the martingales $\mathcal{M}^y$ in (3.4) reaches R in absolute value, Dynkin's formula (3.4) bounds every local time by CR, and hence the elapsed time by CR^{d+1}; Freedman's inequality then shows that the martingales are unlikely to reach R before the walk reaches $\{\Psi \geq R\}$. When the walk reaches $\{\Psi \geq R\}$, the potential at the contact point (Section 4.2) shows that the local times are small at the sites outside the largest ball of Ψ about the origin contained in the cells of the range. The walk has reached $\{\Psi \geq R\}$ by crossing these sites against the drift, so the crossing estimate of Section 4.1 shows that the radius of this ball has order R. On the ball of half this radius the local times then have order R, so the elapsed time is at least cR^{d+1}, and the coarse bounds at time n follow by taking for R a large multiple of r_n.

Proposition 4.1. *Let $p > 0$. There exist $c(d, \kappa, \Psi, p) > 0$ and $C(d, \kappa, \Psi, p) < \infty$ such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$,*

$$cr_n^d \leq |A_n| \leq Cr_n^d, \qquad cr_n \leq \max_{x \in \mathbb{Z}^d} \ell_n(x) \leq Cr_n, \qquad cr_n \leq R_{\mathrm{out}}(n) \leq Cr_n\,.$$

4.1. Crossing a region with small local times.

Outside a ball $\{\Psi \leq a\}$, the drift at every first departure points toward the ball, by convexity. If the local times outside the ball are at most L, then each site there with positive local time has at most L departures, exactly one of them a first departure. We use the exponential of the distance to the ball, with rate of order $1/(L+1)$, and group its conditional increments by site: far from the ball, the drift of the first departure from a site outweighs the second-order terms of all the departures from it. So the walk is unlikely to get farther than order $(L+1)\log N$ from the ball by time N.

For every $p > 0$ there exists $C(d, \kappa, \Psi, p) < \infty$ such that, for every integer $N \geq 2$, with probability at least $1 - CN^{-p}$, simultaneously for all integers $0 \leq t \leq N$ and all real $a \geq 0$,

$$\max_{0 \leq j \leq t} \operatorname{dist}(X_j, \{\Psi \leq a\}) \leq C\Big(1 + \max_{x \in \mathbb{Z}^d, \Psi(x) \geq a} \ell_t(x)\Big) \log N\,. \tag{4.1}$$

Proof of (4.1). Fix $a \geq 0$, and let $\delta(x)$ be the Euclidean distance from x to $\{\Psi \leq a\}$. If $\delta(x) \geq 1$, let $\bar{x}$ be the point of $\{\Psi \leq a\}$ nearest to x, and let $\nu(x) \coloneqq (x - \bar{x})/\delta(x)$. Bounding $\delta(x+e)$ by $|x + e - \bar{x}|$ gives $\delta(x+e) \leq \delta(x) + \nu(x) \cdot e + 1/(2\delta(x))$ for every unit vector $e \in \mathbb{Z}^d$. The subgradient inequality gives $\xi(x) \cdot (x - \bar{x}) \geq \Psi(x) - a$, and the point $ax/\Psi(x)$ of the ball gives $\delta(x) \leq (\Psi(x) - a)/c_\Psi$; so $\xi(x) \cdot \nu(x) \geq c_\Psi$. Let $0 < \theta \leq 1$ and $f \coloneqq e^{\theta\delta}$. Since δ is 1-Lipschitz and $e^u \leq 1 + u + u^2$ for $|u| \leq 1$, at every time j with $\delta(X_j) \geq 1$,

$$\mathbb{E}\big(f(X_{j+1}) - f(X_j) \,\big|\, \mathcal{F}_j\big) \leq \theta f(X_j)\Big(\theta + \frac{1}{2\delta(X_j)} - \kappa c_\Psi I_j\Big)\,.$$

Fix an integer $L \ge 0$ and a real $s > 0$, and let $\theta := \kappa c_\Psi/(2L+2)$. Let T be the first time $t \le N$ at which $\delta(X_t) \ge s$ or some site x with $\Psi(x) \ge a$ has $\ell_t(x) = L+1$, and $T := N$ if there is none. A site x with $\delta(x) \ge (L+1)/(\kappa c_\Psi)$, a distance larger than 1 by (2.2), has at most $L+1$ departures before T, and, if it has any, exactly one of them is a first departure. So the bounds above at these departures sum to at most

$$\theta f(x)\Big((L+1)\Big(\theta + \frac{1}{2\delta(x)}\Big) - \kappa c_\Psi\Big) \le 0\,.$$

At the other sites and their neighbors, $\theta\delta \le 1$, so each departure from these sites contributes at most e. Since $f(X_0) = 1$, it follows that $\mathbb{E}f(X_T) \le 1 + eN$. On the event that $\max_{j\le t}\delta(X_j) \ge s$ and $\max_{\Psi(x)\ge a}\ell_t(x) \le L$ for some $t \le N$, the time T is the first time at which the walk is at distance at least s from the ball, so $f(X_T) \ge e^{\theta s}$. By Markov's inequality, this event has probability at most $(1+eN)e^{-\theta s} \le CN^{-p-2}$ when $s = C(L+1)\log N$ with C large. A union bound over the integers $0 \le L \le N$ and $0 \le a \le \Lambda_\Psi N$ proves (4.1) for these a, because the local times at time $t \le N$ are at most N; for integers $a > \Lambda_\Psi N$ the path lies in the ball. For real a, apply the bound with $\lceil a\rceil$ in place of a: this does not increase the maximum of the local times, and, by radial scaling, every point of $\{\Psi \le \lceil a\rceil\}$ lies within distance $1/c_\Psi$ of $\{\Psi \le a\}$. □

Since $\{\Psi \le a\} \subset B(0, a/c_\Psi)$, the same event gives, for all $t \le N$ and $a \ge 0$,

$$R_{\text{out}}(t) \le \frac{a}{c_\Psi} + C\Big(1 + \max_{x\in\mathbb{Z}^d, \Psi(x)\ge a} \ell_t(x)\Big)\log N\,. \tag{4.2}$$

For the Euclidean norm, $c_\Psi = 1$ and the maximum is over the sites with $|x| \ge a$.

4.2. **The largest ball of the norm in the range.** The following objects are used in Sections 4–7. Let

$$b := \inf\{\Psi(y) : y \in \mathbb{R}^d \setminus D_n\} \qquad \text{and} \qquad E := D_n \setminus \{\Psi < b\}\,,$$

so that $\{\Psi < b\}$ is the largest ball of Ψ about the origin contained in D_n. Choose a point $y_0 \in \overline{\mathbb{R}^d \setminus D_n}$ with $\Psi(y_0) = b$, the **contact point**, and a cell C_z disjoint from D_n whose closure contains y_0. Then $\ell_n(z) = 0$ and $\Psi(z) \ge b$. Let

$$H := U_{D_n}(y_0) \qquad \text{and} \qquad \mathcal{I} := \int_E (\Psi(v) - b)\,\mathrm{d}v\,.$$

Lemma 2.2 gives $U_{\{\Psi<b\}}(y_0) = 0$, so $H = U_E(y_0)$. For almost every $v \in E$, (2.3) for the subgradient $\nabla\Psi(v)$ gives $\nabla\Psi(v)\cdot(v - y_0) \ge \Psi(v) - b \ge 0$. Hence $H \ge 0$, and, if $E \subset B(y_0, \rho)$ for some $\rho > 0$, then

$$H \ge \frac{2\kappa}{\omega_d}\int_E \frac{\Psi(v) - b}{|v - y_0|^d}\,\mathrm{d}v \ge \frac{c\mathcal{I}}{\rho^d}\,, \qquad \text{so} \quad \mathcal{I} \le C\rho^d H\,. \tag{4.3}$$

Since $\widetilde{\ell}_n$ vanishes on C_z and U_{D_n} is continuous,

$$0 \le H \le \sup_{y\in C_z} \big|\widetilde{\ell}_n(y) - U_{D_n}(y)\big|\,. \tag{4.4}$$

The integral $\mathcal{I}$ also bounds the potential of E: splitting E at the level $\Psi = b + a$, Lemma 3.4 and (3.11) bound the potentials of the two parts by $2d\kappa a$ and $C(\mathcal{I}/a)^{1/d}$, and the choice $a := \mathcal{I}^{1/(d+1)}$ gives

$$\sup_{y\in\mathbb{R}^d} |U_E(y)| \le C\mathcal{I}^{1/(d+1)}\,; \tag{4.5}$$

if $\mathcal{I} = 0$, then $|E| = 0$.

4.3. **Proof of the coarse bounds.**

Proof of Proposition 4.1. For $R \geq 2$, let $\tau_R := \inf\{t : \Psi(X_t) \geq R\}$. We prove that, for every $p > 0$, with probability at least $1 - CR^{-p}$,

$$cR^{d+1} \leq \tau_R \leq CR^{d+1} \qquad \text{and} \qquad \max_x \ell_{\tau_R}(x) \leq CR\,.$$

This implies the proposition: for $R = C_1 r_n$ with $cR^{d+1} > n$ we get $n < \tau_R$, hence $R_{\text{out}}(n) \leq Cr_n$ and $\max_x \ell_n(x) \leq Cr_n$, and the remaining bounds follow from $n \leq |A_n| \max_x \ell_n(x)$ and $|A_n| \leq C(1 + R_{\text{out}}(n))^d$.

Let C_0 be so large that $B(0, C_0R/2)$ contains every cell within distance $\sqrt{d}$ of $\{\Psi < R\}$. Before τ_R the walk departs only from sites in $\{\Psi < R\}$, so (3.4) and (3.3) give $\ell_t(y) \leq |\mathcal{M}_t^y| + CR$ for $t \leq \tau_R$. Stop the martingales $\mathcal{M}^y$, $y \in \mathbb{Z}^d \cap B(0, C_0R)$, at τ_R and when one of them first reaches R in absolute value. Until then every local time is at most CR, so the elapsed time is at most CR^{d+1} and, by (3.5), every stopped bracket is at most $CR \log R$. Freedman's inequality [Freedman, 1975, Theorem 1.6] and a union bound over the $O(R^d)$ sites y show that, except with probability CR^{-p}, the stopped martingales stay below $C\sqrt{R \log R} < R$. On this event, the threshold is not reached before τ_R, which gives the upper bounds.

For the lower bound we work on this event and take R large. Let $N := \lceil CR^{d+1} \rceil$, with the constant of the upper bound on τ_R. Except with probability CR^{-p}, the event of (4.1) also holds, so (4.2) holds at time $t = \tau_R \leq N$, with $\log N \leq C \log R$. The comparison (3.9) at time τ_R, whose error terms depend only on the departed sites, and the martingale bound give $|\widetilde{\ell}_{\tau_R} - U_{D_{\tau_R}}| \leq C\sqrt{R \log R}$ on $B(0, C_0R/2)$. For the objects of Section 4.2 at time τ_R, the bounds (4.4), (4.3) with $\rho = C_0R$ and (4.5) give $H \leq C\sqrt{R \log R}$, $\mathcal{I} \leq CR^d H$ and $\sup |U_E| \leq C(R^{d+1/2} \log R)^{1/(d+1)} = o(R/\log R)$. Since $U_{D_{\tau_R}} = 2d\kappa(b - \Psi)_+ + U_E$ by Lemma 2.2, every site of A_{τ_R} outside $\{\Psi < b\}$ has local time $o(R/\log R)$, and (4.2) with $a = b$ gives $R/\Lambda_\Psi \leq R_{\text{out}}(\tau_R) \leq b/c_\Psi + o(R)$, so $b \geq cR$. Then $\widetilde{\ell}_{\tau_R} \geq d\kappa b - o(R)$ on $\{\Psi \leq b/2\}$, and integrating gives $\tau_R \geq cR^{d+1}$. □

5. The limit shape

We prove Theorem 2.1 with rates (Proposition 5.1). Let b, y_0 and C_z be as in Section 4.2, so that $\{\Psi < b\}$ is the largest ball of Ψ about the origin contained in D_n and y_0 is the contact point. By convexity, every point v of D_n outside this ball contributes at least $c(\Psi(v) - b)r_n^{-d}$ to $U_{D_n}(y_0)$. With the coarse bounds, Dynkin's formula at the center z of the cell C_z, a site outside A_n, shows that $U_{D_n}(y_0)$ is at most $C\sqrt{r_n \log n}$ if $d = 2$ and at most $C \log n$ if $d \geq 3$ (Lemma 5.2). So little of D_n lies outside the ball, and the martingale obtained from $|X_n|^2$ in (5.5) shows that b is close to r_n.

For the outer bound, the local times outside $\{\Psi < r_n\}$ are then small, and the crossing estimate (4.1) shows that no visited site lies far from the ball $\{\Psi \leq r_n\}$.

Proposition 5.1. *Let Ψ be a norm, let $\xi(x) \in \partial\Psi(x)$ be subgradients chosen at the sites $x \in \mathbb{Z}^d \setminus \{0\}$, and let $\kappa > 0$ satisfy (2.2). For every $p > 0$ there exists $C(d, \kappa, \Psi, p) < \infty$, not depending on the choice of the subgradients, such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$ the following hold simultaneously.*

(i) Radii*:*

$$\Big|\inf_{y \notin D_n} \Psi(y) - r_n\Big| \leq C \begin{cases} r_n^{3/4}(\log n)^{1/4}, & d = 2, \\ (r_n \log n)^{1/2}, & d \geq 3 \end{cases} \tag{5.1}$$

and

$$\max_{0 \leq j \leq n} \Psi(X_j) - r_n \leq C \begin{cases} r_n^{5/6}(\log n)^{7/6}, & d = 2, \\ r_n^{d/(d+1)}(\log n)^{(d+2)/(d+1)}, & d \geq 3 \end{cases}. \tag{5.2}$$

(ii) Local times:

$$\max_{x\in\mathbb{Z}^d} |\ell_n(x) - 2d\kappa(r_n - \Psi(x))_+| \le C \begin{cases} r_n^{5/6}(\log n)^{1/6}, & d = 2, \\ r_n^{d/(d+1)}(\log n)^{1/(d+1)}, & d \ge 3 \end{cases}. \tag{5.3}$$

Throughout the rest of the paper let

$$q_n := \begin{cases} \sqrt{\dfrac{\log n}{r_n}}, & d = 2, \\ \dfrac{\log n}{r_n}, & d \ge 3 \end{cases}. \tag{5.4}$$

The bounds in (5.1), (5.2) and (5.3) are $Cr_n q_n^{1/2}$, $Cr_n q_n^{1/(d+1)} \log n$ and $Cr_n q_n^{1/(d+1)}$.

5.1. **Martingale estimates.** We collect on one event the martingale bounds used in Sections 5–7. Fix $p > 0$, and let K be a constant with $R_{\text{out}}(n) \le (K-1)r_n$ on the event of Proposition 4.1, so that $D_n \subset B(0, Kr_n)$ there for large n. Since every step has squared length one and conditional mean $-\kappa I_j \xi(X_j)$ given $\mathcal{F}_j$, the process

$$\mathcal{Q}_t := \sum_{j<t} 2X_j \cdot \big(X_{j+1} - X_j + \kappa I_j \xi(X_j)\big)$$

is a martingale, and Euler's relation gives

$$|X_n|^2 = n - 2\kappa \sum_{x\in A_n} \Psi(x) + \mathcal{Q}_n\,. \tag{5.5}$$

Let $\widehat{\mathcal{Q}}$ be $\mathcal{Q}$ stopped when the walk leaves $B(0, Kr_n)$. Its increments are at most Cr_n, so by Azuma's inequality $|\widehat{\mathcal{Q}}_n| \le Cr_n\sqrt{n\log n}$ with probability at least $1 - Cn^{-p}$, and on the event of Proposition 4.1 $\widehat{\mathcal{Q}}_n = \mathcal{Q}_n$, so that

$$|\mathcal{Q}_n| \le Cr_n\sqrt{n\log n}\,. \tag{5.6}$$

For $y \in \mathbb{Z}^d$ let N^y be the martingale $\mathcal{M}^f$ for $f = (g_y - \gamma)^2$, where $\gamma := 0$ if $d \ge 3$ and $\gamma := (2/\pi)\log r_n$ if $d = 2$. If $|y| \le 3n$, the increments of N^y through time n are at most $C\bar{g}$, where $\bar{g} := 1$ if $d \ge 3$ and $\bar{g} := \log n$ if $d = 2$, and Lemma 3.2 applies to $N^y/(C\bar{g})$; we take a union bound over these y.

The event of Section 5.1 is the intersection of the events of Proposition 4.1 and Lemma 3.1 with those on which these bounds on $\widehat{\mathcal{Q}}_n$ and N_n^y, (3.7), and (4.1) with $N = n$ hold. Its complement has probability at most Cn^{-p}, and the arguments of Sections 5–7 are deterministic on it. On it, $\max_x \ell_n(x) \le Cr_n$, so (3.1) gives, for $y \in \mathbb{R}^d$ with $|y| \le 2n$,

$$|\widetilde{\ell}_n(y) - U_{D_n}(y)| \le C \begin{cases} \sqrt{r_n}\log n, & d = 2, \\ \sqrt{r_n \log n}, & d \ge 3 \end{cases}. \tag{5.7}$$

5.2. **The radii and the local times.** Let b, E, y_0, z, H and $\mathcal{I}$ be as in Section 4.2. Then $|v - y_0| \le Cr_n$ for $v \in E$, because $D_n \subset B(0, Kr_n)$ and $\Psi(y_0) = b \le Cr_n$, so (4.3) gives $\mathcal{I} \le Cr_n^d H$. By (4.4) and (5.7), $H \le Cr_n^{1/2}\log n$.

Lemma 5.2 improves this bound, because the walk has not departed from z: up to $C\log n$, Dynkin's formula at z bounds H by $|\mathcal{M}_n^z|$. The bracket of $\mathcal{M}^z$ is at most a sum of the local times weighted by the squared increments of g_z. Dynkin's formula for the square of g_z, shifted by the constant γ in the plane, expresses this sum through $\ell_n(z) = 0$, endpoint terms, one term for each first departure, and the martingale N^z, whose bracket is controlled by the same sum. The same argument at every site y bounds $|\ell_n(y) - U_{D_n}(y)|$ in terms of $\ell_n(y)$; we use this bound again in Sections 6 and 7.

Lemma 5.2. *There exists $C(d,\kappa,\Psi,p)<\infty$ such that, on the event of Section 5.1, for large n,*

$$H \le Cr_nq_n = C\begin{cases}\sqrt{r_n\log n}, & d=2,\\ \log n, & d\ge 3\end{cases}.$$

Proof. Let $y\in\mathbb{Z}^d$ with $|y|\le 3n$, and let

$$\mathcal{E}_y := \frac{1}{2d}\sum_{x\in\mathbb{Z}^d}\ell_n(x)\sum_{|e|=1}\big(g_y(x+e)-g_y(x)\big)^2 \ge \frac{\langle \mathcal{M}^y\rangle_n}{2d}.$$

Since $\Delta g_y = \mathbf{1}_{\{y\}}$, the Laplacian of $f_y := (g_y-\gamma)^2$ at x is $(2d)^{-1}\sum_{|e|=1}(g_y(x+e)-g_y(x))^2 + 2(g(0)-\gamma)\mathbf{1}_{\{y\}}(x)$, so Dynkin's formula for f_y, grouped by site as in (3.4), gives

$$\mathcal{E}_y = 2\big(\gamma-g(0)\big)\ell_n(y) + f_y(X_n) - f_y(0) - N_n^y + \kappa\sum_{x\in A_n}\xi(x)\cdot\bar\nabla f_y(x).$$

Within distance $4n$ of y, $|g_y-\gamma|\le C\bar g$, so $\langle N^y\rangle_n \le C\bar g^2\mathcal{E}_y$ and the first three terms are $O(\bar g\ell_n(y)+\bar g^2)$. By (3.2) and (3.3), the term of x in the last sum is $O((1+|x-y|)^{3-2d})$ if $d\ge 3$ and $O\big((1+|x-y|)^{-1}(1+|\log((1+|x-y|)/r_n)|)\big)$ if $d=2$. These bounds decrease in $|x-y|$, so their sum over the at most Cr_n^d sites of A_n is $O(\log n)$ if $d\ge 3$ and $O\big(\int_1^{Cr_n}(1+|\log(s/r_n)|)\,\mathrm{d}s\big)=O(r_n)$ if $d=2$. Here the shift γ matters: without it the planar sum would be of order $r_n\log r_n$. The bound of Lemma 3.2 for N^y and Young's inequality then give, since $(\log n)^3 = o(r_n)$,

$$\mathcal{E}_y \le C\begin{cases}\ell_n(y)\log n + r_n, & d=2,\\ \ell_n(y)+\log n, & d\ge 3\end{cases},$$

and with (3.7) and (3.9), simultaneously for all $y\in\mathbb{Z}^d$ with $|y|\le 3n$,

$$|\ell_n(y)-U_{D_n}(y)| \le C\begin{cases}\sqrt{r_n\log n}+(\log n)\sqrt{\ell_n(y)}, & d=2,\\ \log n+\sqrt{\ell_n(y)\log n}, & d\ge 3\end{cases}. \tag{5.8}$$

Young's inequality absorbs half of $\ell_n(y)$ on the right side, because $(\log n)^2 = o(\sqrt{r_n\log n})$ when $d=2$; so $U_{D_n}(y)\ge \ell_n(y)/2 - Cr_nq_n$, and $|U_{D_n}(y)|\le Cr_nq_n$ if $y\notin A_n$. By (3.10), and since $\widetilde{\ell}_n$ vanishes and $|U_{D_n}|\le C$ outside $B(0,2n)$, it follows that

$$U_{D_n} \ge \frac{1}{2}\widetilde{\ell}_n - Cr_nq_n \quad\text{on } \mathbb{R}^d \qquad\text{and}\qquad |U_{D_n}|\le Cr_nq_n \quad\text{on } \mathbb{R}^d\setminus D_n. \tag{5.9}$$

Since y_0 lies in the closure of C_z, which is disjoint from D_n, continuity of U_{D_n} gives $H\le Cr_nq_n$. □

Proof of Proposition 5.1. We work on the event of Section 5.1 and take n large. By Lemma 5.2, $\mathcal{I}\le Cr_n^dH\le Cr_n^{d+1}q_n$. Splitting E at the level $\Psi = b+a$ with $a := r_nq_n^{1/2}$, into a shell of volume $O(r_n^{d-1}a)$ and a set of volume at most $\mathcal{I}/a$, gives $|E|\le Cr_n^dq_n^{1/2}$. By (5.5), (5.6) and $|X_n|\le Cr_n$, $n-2\kappa\sum_{x\in A_n}\Psi(x) = O(r_n^{(d+3)/2}(\log n)^{1/2})$; the sum differs from $\int_{D_n}\Psi$ by $O(r_n^d)$, and $\int_{D_n}\Psi = d|B_\Psi|b^{d+1}/(d+1)+O(r_n|E|)$. Dividing by $n = 2d\kappa|B_\Psi|r_n^{d+1}/(d+1)$ gives

$$\left|1-\left(\frac{b}{r_n}\right)^{d+1}\right| \le C\big(r_n^{(1-d)/2}(\log n)^{1/2} + r_n^{-1} + |E|r_n^{-d}\big), \tag{5.10}$$

which is at most $Cq_n^{1/2}$; so $|b-r_n|\le Cr_nq_n^{1/2}$, which is (5.1). Discarding the nonnegative integral over E and the term $-|X_n|^2$ in (5.5) also gives $(b-r_n)_+\le C(r_n^{(3-d)/2}(\log n)^{1/2}+1)$. By Lemma 2.2, $U_{D_n} = 2d\kappa(b-\Psi)_+ + U_E$, and (4.5) gives $\sup|U_E|\le C(r_n^{d+1}q_n)^{1/(d+1)} = Cr_nq_n^{1/(d+1)}$. With (5.7) and the bound on $|b-r_n|$, this gives $|\widetilde{\ell}_n - 2d\kappa(r_n-\Psi)_+|\le Cr_nq_n^{1/(d+1)}$ on $\mathbb{R}^d$, which is (5.3). Finally, this

bound gives $\ell_n(x) \le C r_n q_n^{1/(d+1)}$ at the sites with $\Psi(x) \ge r_n$, so (4.1) with $a = r_n$ and $t = N = n$, together with $\Psi(x) - r_n \le \Lambda_\Psi \operatorname{dist}(x, \{\Psi \le r_n\})$, proves (5.2). □

Proof of Theorem 2.1. Apply Proposition 5.1 with $p > 1$ and the Borel–Cantelli lemma. Its errors are $o(r_n)$, so the inner- and outer-radius bounds give (i), and the local-time bound gives (ii). At every fixed site x, part (ii) gives $\ell_n(x)/r_n \to 2d\kappa > 0$, proving (iii). Taking the Euclidean norm gives Theorem 1.1. □

The proofs in Sections 2–5 use the symmetry $\Psi(-x) = \Psi(x)$ only to bound the coordinates of the subgradients. Let $B \subset \mathbb{R}^d$ be a compact convex set with the origin in its interior, and let $\Psi_B(x) := \inf\{t > 0 : x \in tB\}$ be its Minkowski functional, which is convex and positively homogeneous and satisfies $c|x| \le \Psi_B(x) \le C|x|$. Every $\xi \in \partial\Psi_B(x)$ satisfies $-\Psi_B(-e_i) \le \xi_i \le \Psi_B(e_i)$; so, if $\kappa \max\{\Psi_B(e_i), \Psi_B(-e_i)\} < 1/d$ for $1 \le i \le d$, then the probabilities (2.1) are positive, and the proofs of Theorem 2.1 and Proposition 5.1 apply without change to Ψ_B, with $|B|$ in place of $|B_\Psi|$. Thus every such set B is the limit shape of the centrally excited random walk with Ψ_B in place of Ψ.

6. The inner radius for the Euclidean norm

We prove the bounds on $|R_{\text{in}}(n) - r_n|$ and on the volume in Theorem 1.2 (i) and (ii), and a first uniform bound on the local times (Proposition 6.1). For the Euclidean norm the contact point y_0 lies on the sphere $\{|v| = R_{\text{in}}(n)\}$, and every point v of D_n outside the ball $B(0, R_{\text{in}}(n))$ contributes at least $c|v - y_0|^{2-d}/r_n$ to the potential at y_0 (inequality (6.2) below and Figure 5). Unlike the lower bound $c(\Psi(v) - b)r_n^{-d}$ of Section 5, this bound does not vanish near the sphere, so Lemma 5.2 bounds the volume of D_n outside the ball, and not only the integral $\mathcal{I}$.

Proposition 6.1 (Inner radius)**.** *Let Ψ be the Euclidean norm, and let $0 < \kappa < 1/d$. For every $p > 0$ there exists $C(d, \kappa, p) < \infty$ such that, for every integer $n \ge 2$, with probability at least $1 - Cn^{-p}$, the following hold simultaneously, where q_n is defined in (5.4).*

(i) Inner radius*:*

$$|R_{\text{in}}(n) - r_n| \le C r_n q_n \,. \tag{6.1}$$

(ii) Volume*:*

$$\bigl|r_n^{-1} D_n \mathbin{\triangle} B(0,1)\bigr| \le C q_n \,.$$

(iii) Local times*:*

$$\sup_{y \in \mathbb{R}^d} \bigl|\widetilde{\ell}_n(y) - 2d\kappa(r_n - |y|)_+\bigr| \le C r_n q_n^{1/d} \,.$$

Fix $p > 0$. In this section and the next we take n large and work on the event of Section 5.1; its complement has probability at most Cn^{-p}, and on it $D_n \subset B(0, Kr_n)$, with K as in Section 5.1. Let $b := R_{\text{in}}(n)$ and $E := D_n \setminus B(0, b)$, and let y_0 and H be as in Section 4.2 for the Euclidean norm, so that $|y_0| = b$ and $H = U_{D_n}(y_0) = U_E(y_0)$.

If $s = |v| \ge b = |y_0|$, then

$$\frac{u_v \cdot (v - y_0)}{|v - y_0|^d} = \frac{s^2 - v \cdot y_0}{s|v - y_0|^d} \ge \frac{1}{2s}|v - y_0|^{2-d} \ge 2^{1-d} s^{1-d} \,. \tag{6.2}$$

The first inequality uses $2(s^2 - v \cdot y_0) = |v - y_0|^2 + s^2 - b^2$, and the second $|v - y_0| \le 2s$ and $d \ge 2$. Since $|v| \le C r_n$ on E, integrating (6.2) over E gives $H \ge c|E| r_n^{1-d}$, and Lemma 5.2 gives

$$|E| \le C r_n^{d-1} H \le C r_n^d q_n \,. \tag{6.3}$$

Proof of Proposition 6.1. Substituting (6.3) into (5.10) gives $|b - r_n| \le C r_n q_n$, which is (i), and with (6.3) also (ii). On $B(0, Kr_n)$, the decomposition $U_{D_n}(y) = 2d\kappa(b - |y|)_+ + U_E(y)$, the volume bound (3.11), (5.7) and (i) give

$$\bigl|\widetilde{\ell}_n(y) - 2d\kappa(r_n - |y|)_+\bigr| \le C\bigl(\sqrt{r_n}\log n + r_n q_n + |E|^{1/d}\bigr) \le C r_n q_n^{1/d} \,,$$

and outside this ball both terms on the left vanish; this proves (iii). □

7. The outer radius and the local times

We prove the outer-radius and local-time bounds in Theorem 1.2. The crossing estimate (4.2) bounds the difference of the radii by $C \log n$ times one plus the largest local time outside the inner ball. To bound these local times, we bound the volume of the set $E = D_n \setminus B(0, R_{\text{in}}(n))$ near the outer sphere, the sphere about the origin of radius $R_{\text{out}}(n) + 2\sqrt{d}$. At a point beyond this sphere the potential is small, because the local time vanishes there. The points of E near that point contribute negatively to the potential, so a bound on the positive contributions bounds their volume (Figure 5). Because the outer sphere is curved, the positive contributions are bounded by the integral of $|v-y|^{2-d}/|v|$ over E. We first bound this integral by Cr_nq_n, using the volume bound in the plane and integrating the potential along rays in higher dimensions.

Proposition 7.1 (Outer radius)**.** *Let Ψ be the Euclidean norm, and let $0 < \kappa < 1/d$. For every $p > 0$ there exists $C(d, \kappa, p) < \infty$ such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$,*

$$R_{\text{out}}(n) \leq r_n + C \begin{cases} \sqrt{r_n}(\log n)^{5/2}, & d = 2, \\ (\log n)^{d+1}, & d \geq 3 \end{cases}.$$

With $b = R_{\text{in}}(n)$ and $E = D_n \setminus B(0, b)$, let

$$w := R_{\text{out}}(n) - b + 2\sqrt{d} \qquad \text{and} \qquad L := 1 + \max\{\ell_n(x) : x \in \mathbb{Z}^d,\ |x| \geq b\}.$$

Then $w \geq 3\sqrt{d}/2$, because $b \leq R_{\text{out}}(n) + \sqrt{d}/2$, and $E \subset \{b \leq |v| \leq b + w\}$. By (5.2) and (6.1), $w = o(r_n)$, so $1 \leq w \leq b$ and $b \asymp r_n$ for large n. The crossing estimate (4.2) with $t = N = n$ and $a = b$ gives

$$w \leq CL \log n. \tag{7.1}$$

7.1. The potential beyond the inner radius.

For a bounded measurable set $D \subset \mathbb{R}^d$ and $y \in \mathbb{R}^d$, write

$$U_D^+(y) := \frac{2\kappa}{\omega_d} \int_D \left[u_v \cdot \frac{v - y}{|v - y|^d} \right]_+ \mathrm{d}v$$

for the potential with its negative contributions discarded. If $|y| \geq b + w$, then every $v \in E$ satisfies $|v| \leq |y|$, so the identity $2(|v|^2 - v \cdot y) = |v - y|^2 + |v|^2 - |y|^2$ gives $u_v \cdot (v - y) \leq |v - y|^2/(2|v|)$.

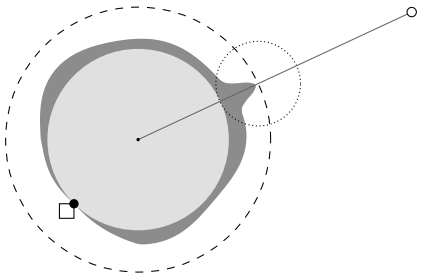


FIGURE 5. The light disk is the largest ball $B(0, b)$ about the origin in D_n, and the dark region is $E = D_n \setminus B(0, b)$. Every point of E contributes nonnegatively to $U_E(y_0)$ at the contact point y_0 (black dot), which lies on the circle $\{|v| = b\}$ and in the closure of a cell disjoint from D_n (square). The dashed circle, of radius $b + w = R_{\text{out}}(n) + 2\sqrt{d}$, contains D_n, and the points of E in the dotted ball $B((b + w)u, t)$ contribute negatively to $U_E(y_t)$ at the point $y_t = (b + w + 4t)u$ (open dot) on the segment from the origin (small dot) in the direction of a unit vector u.

Hence, with

$$F(y) := \int_E \frac{|v-y|^{2-d}}{|v|}\,\mathrm{d}v\,,$$

every $y \in \mathbb{R}^d$ with $|y| \geq b+w$ satisfies

$$U_E^+(y) \leq \frac{\kappa}{\omega_d} F(y)\,. \tag{7.2}$$

We prove that

$$\sup_{\mathbb{R}^d} F \leq C r_n q_n\,. \tag{7.3}$$

Proof of (7.3). If $d = 2$, then $F \leq |E|/b$, and (6.3) gives the bound. If $d \geq 3$, differentiating $t^{d-2}|v-tu|^{2-d}$ in t gives, for every unit vector u,

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(t^{d-2}F(tu)\big) = \frac{(d-2)\omega_d}{2\kappa} t^{d-3} U_E(tu)\,.$$

For $t \geq b$, (5.9) gives $U_E(tu) = U_{D_n}(tu) \geq -C r_n q_n$, and (6.3) gives $F(2(b+w)u) \leq |E|(b+w)^{2-d}/b \leq C r_n q_n$. Integrating over $[t, 2(b+w)]$ gives $t^{d-2}F(tu) \leq C r_n^{d-1} q_n$, hence, since $b \asymp r_n$, the bound on $b \leq |y| \leq 2(b+w)$, and the maximum principle for F, which is harmonic off the closure of E and tends to zero at infinity, extends it to $\mathbb{R}^d$. □

7.2. **The volume near the outer sphere and the outer radius.** The following deterministic lemma bounds the positive contributions to the potential at a point between the two radii in terms of the volume near the sphere of radius $b+w$.

Lemma 7.2. *There exists $C(d,\kappa) < \infty$ with the following property. Let $1 \leq w \leq b$ and $\lambda \geq 1$, and let D be a measurable subset of $\{v \in \mathbb{R}^d : b \leq |v| \leq b+w\}$ such that $|D \cap B((b+w)u, t)| \leq \lambda t^{d-1}$ for every unit vector u and every $t \geq w$, and $\int_D |v-y|^{2-d}\,\mathrm{d}v \leq \lambda b$ for every $y \in \mathbb{R}^d$. Then every $y \in \mathbb{R}^d$ with $b \leq |y| \leq b+w$ satisfies*

$$U_D^+(y) \leq C\big(\lambda w^{d-1}\big)^{1/d} + C\lambda\,.$$

Proof. Since $B(y,t) \subset B((b+w)u_y, 2t)$ for $t \geq w$, the first hypothesis gives $|D \cap B(y,t)| \leq 2^{d-1}\lambda t^{d-1}$. So the points of D within distance w of y contribute at most $C(\lambda w^{d-1})^{1/d}$, as in (3.11). For the other points, the identity $2(|v|^2 - v\cdot y) = |v-y|^2 + |v|^2 - |y|^2$ and $|v|^2 - |y|^2 \leq 3bw$ bound the kernel by $2w|v-y|^{-d} + |v-y|^{2-d}/(2b)$. By the second hypothesis, the second term contributes at most $C\lambda$, and, integrating by parts in $|v-y|$, the first contributes at most $Cw\int_w^\infty \lambda t^{d-1} t^{-d-1}\,\mathrm{d}t \leq C\lambda$. □

Proof of Proposition 7.1. We first show that, for every unit vector u and every $t \geq w$,

$$|E \cap B((b+w)u, t)| \leq C r_n q_n t^{d-1}\,. \tag{7.4}$$

For $t > r_n/4$ this follows from (6.3). For $w \leq t \leq r_n/4$, let $y_t := (b+w+4t)u$ (Figure 5). Then $y_t \notin D_n$, because $R_{\mathrm{out}}(n) = b+w-2\sqrt{d}$, and $U_{B(0,b)}(y_t) = 0$, so (5.9) gives $U_E(y_t) \geq -C r_n q_n$. If $v \in E \cap B((b+w)u, t)$, then $u_v \cdot u \geq 1 - t/(b+w)$ and $t \leq (b+w)/2$, so $u_v \cdot (v - y_t) \leq (b+w) - (b+w+4t)(1-t/(b+w)) \leq -t$, while $|v - y_t| \leq 5t$; so these points contribute at most $-ct^{1-d}$ each to the kernel of $U_E(y_t)$. By (7.2) and (7.3), the other points contribute at most $C r_n q_n$, which proves (7.4).

Since $|v| \leq 2b$ on E, (7.3) gives $\int_E |v-y|^{2-d}\,\mathrm{d}v \leq 2bF(y) \leq Cb r_n q_n$. With (7.4), Lemma 7.2 applies to $D = E$ and $\lambda = C r_n q_n$, and gives, for $b \leq |y| \leq b+w$,

$$U_E^+(y) \leq C\big(r_n q_n w^{d-1}\big)^{1/d} + C r_n q_n\,. \tag{7.5}$$

For $|y| \geq b$, (5.9) gives $\widetilde{\ell}_n(y) \leq 2U_E^+(y) + Cr_nq_n$, so $L \leq C((r_nq_nw^{d-1})^{1/d} + r_nq_n)$. With (7.1) and Young's inequality, $w \leq w/2 + Cr_nq_n(\log n)^d$, that is,

$$w \leq Cr_nq_n(\log n)^d = C\begin{cases}\sqrt{r_n}(\log n)^{5/2}, & d=2,\\ (\log n)^{d+1}, & d\geq 3\end{cases}. \tag{7.6}$$

Since $R_{\mathrm{out}}(n) \leq b + w$ and $|b - r_n| \leq Cr_nq_n$, this proves the proposition. □

7.3. **The uniform bound on the local times.** We prove Theorem 1.2 (iii) on the event of Section 5.1, by bounding U_E on the annulus $\{b \leq |y| \leq b + w\}$, which contains E. There the ball potential vanishes, so (5.9) gives $U_E = U_{D_n} \geq -Cr_nq_n$, and $U_E \leq U_E^+$ and (7.5) give the upper bound. The function U_E is continuous, harmonic off the closure of E, and tends to zero at infinity, so the maximum principle extends both bounds to $B(0, b)$ and to the complement of $B(0, b + w)$. With (7.6), this gives

$$\sup_{\mathbb{R}^d}|U_E| \leq C\big((r_nq_nw^{d-1})^{1/d} + r_nq_n\big) \leq Cr_nq_n(\log n)^{d-1} = C\begin{cases}\sqrt{r_n}(\log n)^{3/2}, & d=2,\\ (\log n)^d, & d\geq 3\end{cases}.$$

On $B(0, Kr_n)$, combine this with (5.7), $U_{D_n}(y) = 2d\kappa(b - |y|)_+ + U_E(y)$ and $|b - r_n| \leq Cr_nq_n$ to obtain the claimed bound. Outside this ball both the local time and $2d\kappa(r_n - |y|)_+$ vanish.

Proof of Theorem 1.2. On the event of Section 5.1, for large n, Proposition 6.1 gives the bounds on $R_{\mathrm{in}}(n)$ and on the volume, Proposition 7.1 the bound on $R_{\mathrm{out}}(n)$, and Section 7.3 the bound on the local times. The Borel–Cantelli lemma with $p > 1$ gives the almost-sure statement. □

8. Limit laws

We prove central limit theorems and laws of the iterated logarithm for $\sum_{x\in A_n}|x|$ and for the local times at fixed sites. After centering, these quantities are, up to smaller errors, linear combinations of the martingale $\mathcal{Q}$ of (5.5) and of the martingales $\mathcal{M}^y$ of Section 3.1; we compute the limits of their brackets from Theorem 1.1 (ii), and then apply the martingale central limit theorem and the law of the iterated logarithm.

We use these two theorems in the following form, here and in Section 9. Let Z be a real square-integrable martingale with $Z_0 = 0$, and let $s_n \to \infty$ be deterministic with $\log\log s_n \sim \log\log n$. Suppose that $\langle Z\rangle_n/s_n^2 \to \sigma^2$ almost surely for a constant $\sigma \geq 0$ and that $|Z_{j+1} - Z_j| \leq \eta_j$ for $\mathcal{F}_j$-measurable η_j with $\max_{j<n}\eta_j = o(s_n(\log\log n)^{-1/2})$ almost surely. Then Z_n/s_n converges in distribution to $\mathcal{N}(0, \sigma^2)$ [Hall and Heyde, 1980, Corollary 3.1], and, if $\sigma > 0$, then $\limsup_{n\to\infty} \pm Z_n/(s_n\sqrt{2\log\log n}) = \sigma$ almost surely for each sign [Stout, 1970, Theorems 1 and 2]. For a vector martingale satisfying these hypotheses in every direction with the same s_n, the Cramér–Wold device gives a joint Gaussian limit.

8.1. **The sum of the distances from the origin.** For the Euclidean norm, (5.5) reads

$$\mathcal{Q}_n = 2\kappa\sum_{x\in A_n}|x| - n + |X_n|^2. \tag{8.1}$$

Let $R_{\mathrm{mom}}(n)$ be the radius of the ball about the origin on which the integral of $|v|$ equals $\sum_{x\in A_n}|x|$, that is,

$$R_{\mathrm{mom}}(n) := \left(\frac{d+1}{d\omega_d}\sum_{x\in A_n}|x|\right)^{1/(d+1)}. \tag{8.2}$$

Theorem 8.1 (Sum of the distances from the origin)**.** *Let Ψ be the Euclidean norm, let $0 < \kappa < 1/d$, and let $R_{\mathrm{mom}}(n)$ be as in (8.2). Then the following hold.*

(i) Central limit theorem: *as* $n \to \infty$,

$$\frac{\sum_{x\in A_n} |x| - n/(2\kappa)}{r_n^{(d+3)/2}} \xrightarrow{\mathrm{d}} \mathcal{N}\Big(0, \frac{2d\omega_d}{\kappa(d+2)(d+3)}\Big)$$

and

$$\frac{R_{\mathrm{mom}}(n) - r_n}{r_n^{(3-d)/2}} \xrightarrow{\mathrm{d}} \mathcal{N}\Big(0, \frac{2}{\kappa d\omega_d(d+2)(d+3)}\Big)\,.$$

(ii) Law of the iterated logarithm: *for each choice of sign, almost surely,*

$$\limsup_{n\to\infty} \frac{\pm\big(\sum_{x\in A_n} |x| - n/(2\kappa)\big)}{r_n^{(d+3)/2}\sqrt{2\log\log n}} = \Big(\frac{2d\omega_d}{\kappa(d+2)(d+3)}\Big)^{1/2},$$

$$\limsup_{n\to\infty} \frac{\pm(R_{\mathrm{mom}}(n) - r_n)}{r_n^{(3-d)/2}\sqrt{2\log\log n}} = \Big(\frac{2}{\kappa d\omega_d(d+2)(d+3)}\Big)^{1/2}. \tag{8.3}$$

In the plane, comparing $R_{\mathrm{mom}}(n)$ with the inner and outer radii gives Theorem 1.3 (ii) and, with a lower bound on the probability that $\mathcal{Q}_n$ deviates by about $(\log n)^{1/2}$ standard deviations (Lemma 9.1), Theorem 1.3 (i).

Proof of Theorem 8.1. Under both (2.1) and the simple random walk transition probabilities, the conditional second-moment matrix of $X_{j+1} - X_j$ is d^{-1} times the identity, so the conditional variance of $q \cdot (X_{j+1} - X_j)$ is $|q|^2/d - \kappa^2 I_j (q \cdot u_{X_j})^2$ for $\mathcal{F}_j$-measurable q. With $q = 2X_j$ this gives

$$\langle \mathcal{Q} \rangle_n = \frac{4}{d} \sum_{x\in\mathbb{Z}^d} \ell_n(x)|x|^2 - 4\kappa^2 \sum_{x\in A_n} |x|^2\,. \tag{8.4}$$

By Theorem 1.1, the second sum is $O(r_n^{d+2})$ and the first is r_n^{d+3} times $2d\kappa \int_{|v|<1} (1-|v|)|v|^2 \,\mathrm{d}v + o(1)$, almost surely, so

$$\frac{\langle \mathcal{Q} \rangle_n}{r_n^{d+3}} \longrightarrow \frac{8d\kappa\omega_d}{(d+2)(d+3)} \qquad \text{almost surely}\,.$$

Since $|\mathcal{Q}_{j+1} - \mathcal{Q}_j| \le 4|X_j| = O(r_n)$, the limit theorems stated at the beginning of this section apply to $\mathcal{Q}$ with $s_n = r_n^{(d+3)/2}$, and the endpoint term $|X_n|^2 = O(r_n^2)$ in (8.1) is negligible; this gives the assertions for the sum. The derivative of $((d+1)s/(d\omega_d))^{1/(d+1)}$ at $s = n/(2\kappa)$ is $1/(d\omega_d r_n^d)$, which gives the assertions for $R_{\mathrm{mom}}(n)$. □

8.2. **The potential at a fixed site.** Since $B(0, R_{\mathrm{in}}(n)) \subset D_n \subset B(0, R_{\mathrm{out}}(n) + \sqrt{d})$, Theorem 1.2 (i) and (ii) give, almost surely for all large n, $|D_n \triangle B(0, r_n)| \le C r_n^d q_n$ and, for every $v \in D_n \triangle B(0, r_n)$,

$$\big||v| - r_n\big| \le C \begin{cases} \sqrt{r_n}(\log n)^{5/2}, & d = 2, \\ (\log n)^{d+1}, & d \ge 3 \end{cases}. \tag{8.5}$$

For a fixed site y, on the annulus defined by (8.5) the kernel of $U_{D_n}(y)$ is close to $r_n^{-d}|v|$, which relates $U_{D_n}(y)$ to $\sum_{x\in A_n} |x|$ and hence, by (8.1), to $\mathcal{Q}_n$.

Lemma 8.2. *Let Ψ be the Euclidean norm, let $0 < \kappa < 1/d$, and let $y \in \mathbb{Z}^d$. There exists $C(d, \kappa, y) < \infty$ such that, almost surely, for all sufficiently large n,*

$$\Big| U_{D_n}(y) - 2d\kappa(r_n - |y|) - \frac{\mathcal{Q}_n}{\omega_d r_n^d} \Big| \le C \begin{cases} (\log n)^3, & d = 2, \\ 1, & d \ge 3 \end{cases}, \tag{8.6}$$

where $\mathcal{Q}$ is the martingale of (5.5).

By Theorem 8.1, $\mathcal{Q}_n/(\omega_d r_n^d)$ has order $r_n^{(3-d)/2}$, smaller than the order of $\mathcal{M}_n^y$ by a factor $(\log r_n)^{1/2}$ in the plane and $r_n^{(d-2)/2}$ when $d \ge 3$.

Proof. Write $r = r_n$ and let w be the right side of (8.5). For fixed y and $v \in D_n \triangle B(0,r)$, the kernel $u_v \cdot (v-y)|v-y|^{-d}$ equals $|v|^{1-d} + O_y(r^{-d}) = r^{-d}|v| + O_y(r^{-d}(1+w))$. Integrating against $\mathbf{1}_{D_n} - \mathbf{1}_{B(0,r)}$, whose L^1 norm is $O(r^d q_n)$, and using (1.10) and $\int_{B(0,r)} |v|\,\mathrm{d}v = n/(2\kappa)$, gives

$$U_{D_n}(y) - 2d\kappa(r - |y|) = \frac{2\kappa}{\omega_d r^d}\left(\int_{D_n} |v|\,\mathrm{d}v - \frac{n}{2\kappa}\right) + O_y\big(q_n(1+w)\big)\,.$$

Replacing the integral by $\sum_{x\in A_n}|x|$ costs $O(r^d)$, so by (8.1) the main term is $\mathcal{Q}_n/(\omega_d r^d) + O(1)$, and $q_n(1+w)$ is $O((\log n)^3)$ in the plane and $O(1)$ in higher dimensions. □

8.3. **Local times at fixed sites.** Up to a smaller correction at first departures, the bracket of $\mathcal{M}^y$ is a sum of local times with weights that decay like $|x-y|^{2-2d}$; in the plane each dyadic annulus about y inside $B(0,r_n)$ contributes order r_n, which gives an extra factor $\log r_n$. For functions $f_1, f_2 \colon \mathbb{Z}^d \to \mathbb{R}$ with bounded nearest-neighbor increments, let

$$\Gamma(f_1,f_2)(x) \coloneqq \frac{1}{2d}\sum_{|e|=1}\big(f_1(x+e)-f_1(x)\big)\big(f_2(x+e)-f_2(x)\big) - \Delta f_1(x)\Delta f_2(x)\,, \tag{8.7}$$

which is the conditional covariance of the increments of $\mathcal{M}^{f_1}$ and $\mathcal{M}^{f_2}$ at a departure from x other than the first. As in Section 3.1, $g_y = g(\cdot - y)$ and G is the Green function.

Theorem 8.3 (Local times at fixed sites)**.** *Let Ψ be the Euclidean norm, and let $0 < \kappa < 1/d$. Then the following hold.*

(i) Central limit theorem*: for every $y \in \mathbb{Z}^d$, as $n \to \infty$,*

$$\begin{aligned} \frac{\ell_n(y) - 2d\kappa(r_n - |y|)}{\sqrt{r_n \log r_n}} &\xrightarrow{\mathrm{d}} \mathcal{N}\Big(0, \frac{16\kappa}{\pi}\Big) && \text{if } d = 2\,,\\ \frac{\ell_n(y) - 2d\kappa(r_n - |y|)}{\sqrt{r_n}} &\xrightarrow{\mathrm{d}} \mathcal{N}\big(0, 2d\kappa(2G(0)-1)\big) && \text{if } d \ge 3\,. \end{aligned} \tag{8.8}$$

(ii) Law of the iterated logarithm*: for every $y \in \mathbb{Z}^d$ and each choice of sign, almost surely,*

$$\begin{aligned} \limsup_{n\to\infty} \frac{\pm(\ell_n(y) - 2d\kappa(r_n - |y|))}{\sqrt{2r_n \log r_n \log\log n}} &= \Big(\frac{16\kappa}{\pi}\Big)^{1/2} && \text{if } d = 2\,,\\ \limsup_{n\to\infty} \frac{\pm(\ell_n(y) - 2d\kappa(r_n - |y|))}{\sqrt{2r_n \log\log n}} &= \big(2d\kappa(2G(0)-1)\big)^{1/2} && \text{if } d \ge 3\,. \end{aligned}$$

(iii) Joint limits*: for all $y_1, \dots, y_k \in \mathbb{Z}^d$, the convergence in (8.8) holds jointly for $y = y_1, \dots, y_k$, and the limit is a centered Gaussian vector whose covariance matrix has (i,j) entry*

$$\begin{cases} 16\kappa/\pi, & d = 2,\\ 2d\kappa\big(2G(y_i - y_j) - \mathbf{1}_{\{y_i = y_j\}}\big), & d \ge 3 \end{cases}. \tag{8.9}$$

In the plane the limits at different sites are equal; this statement concerns finitely many fixed sites, whereas the maximum in Theorem 1.3 (iii) is over the sites x with $|x| \le r_n^{1/2}$, a set that grows with n.

Proof of Theorem 8.3. A first departure from x changes the conditional covariance from $\Gamma(g_y, g_z)(x)$ by $O_{y,z}((1+|x|)^{2-2d})$, by (3.3), and $A_n \subset B(0, Cr_n)$ eventually almost surely; thus

$$\langle \mathcal{M}^y, \mathcal{M}^z\rangle_n = \sum_{x\in\mathbb{Z}^d} \ell_n(x)\Gamma(g_y,g_z)(x) + \begin{cases} O_{y,z}(\log r_n), & d = 2,\\ O_{y,z}(1), & d \ge 3 \end{cases}. \tag{8.10}$$

The identity $\Gamma(f,h) = \Delta(fh) - f\Delta h - h\Delta f - \Delta f \Delta h$ and summation by parts give

$$\sum_x \Gamma(f,h)(x) = -\sum_x \bigl(f\Delta h + h\Delta f + \Delta f \Delta h\bigr)(x) \tag{8.11}$$

when the boundary terms vanish. If $d \geq 3$, they vanish for $f = g_y$ and $h = g_z$ by (3.2) and (3.3), and since $\Delta g_y = \mathbf{1}_{\{y\}}$ the sum is $2G(y-z) - \mathbf{1}_{\{y=z\}}$. The weights are summable, so Theorem 1.1 (ii) and dominated convergence give $\langle \mathcal{M}^y, \mathcal{M}^z\rangle_n / r_n \to 2d\kappa(2G(y-z) - \mathbf{1}_{\{y=z\}})$ almost surely. If $d = 2$, then $\Gamma(g_y, g_z)(x) = 2/(\pi^2|x|^2) + O_{y,z}(|x|^{-3})$, so $\sum_{|x|\leq r} \Gamma(g_y,g_z)(x) = (4/\pi)\log r + O_{y,z}(1)$. In (8.10) the term $4\kappa r_n$ of the profile $4\kappa(r_n - |x|)_+$ contributes $(16\kappa/\pi) r_n \log r_n + O(r_n)$, the term $-4\kappa|x|$ contributes $O(r_n)$, and the error $o(r_n)$ of Theorem 1.1 (ii) contributes $o(r_n \log r_n)$; so $\langle \mathcal{M}^y, \mathcal{M}^z\rangle_n / (r_n \log r_n) \to 16\kappa/\pi$ almost surely.

The martingales $\mathcal{M}^y$ have bounded increments, so the limit theorems stated at the beginning of this section apply to $(-\mathcal{M}^{y_i})_{i\leq k}$, with $s_n = \sqrt{r_n \log r_n}$ if $d = 2$ and $s_n = \sqrt{r_n}$ if $d \geq 3$. By (3.9) and Lemma 8.2, $\ell_n(y) - 2d\kappa(r_n - |y|) = -\mathcal{M}^y_n + \mathcal{Q}_n/(\omega_d r_n^d) + O_y((\log n)^3)$, and Theorem 8.1 makes the last two terms negligible. □

9. Lower bounds on the fluctuations

We prove the lower bounds of Theorem 1.3, an upper bound on the deviations of the local times away from the boundary (Proposition 9.2), and logarithmic lower bounds on the radii (Proposition 9.4). The lower bounds in Theorem 1.3 (i), (iii) and (iv)(a) need martingale deviations of about $(\log n)^{1/2}$ standard deviations. We obtain them from exponential martingales (Lemma 9.1): for parts (i) and (iv)(a) by the Paley–Zygmund inequality for one exponential martingale, and for part (iii) by a second-moment bound for the average of the exponential martingales at many well-separated sites. Part (iii) needs many sites because, by Theorem 8.3, the deviation at one fixed site is typically smaller than the bound in part (iii) by a factor of order $(\log n)^{1/2}$.

Throughout this section $0 < \kappa < 1/d$, and Ω_n is the event of Section 5.1 with $p = 10$, so $\mathbb{P}(\Omega_n^c) = O(n^{-10})$ and the estimates of Sections 3–7 used below hold on Ω_n for large n. In particular, by Section 7.3, on Ω_n for large n, in every dimension,

$$\max_{x\in\mathbb{Z}^d} \bigl|\ell_n(x) - 2d\kappa(r_n - |x|)_+\bigr| \leq C\sqrt{r_n}(\log n)^{3/2}\,. \tag{9.1}$$

Lemma 9.1. *Let $0 < c_0 \leq C_0$. There exist $c(c_0,C_0) > 0$ and $C(c_0,C_0) < \infty$ with the following property. Let m and n be positive integers, let $b > 0$, $\delta \geq 0$ and $0 \leq \alpha \leq 1$, and let $S^1, \dots, S^m$ be real martingales with respect to a common filtration, with $S^i_0 = 0$ and $|S^i_{t+1} - S^i_t| \leq b$ for $0 \leq t < n$. Let Ω be an event with $\mathbb{P}(\Omega) \geq 1 - \alpha$ on which $c_0 \leq \langle S^i\rangle_n \leq C_0$ for $1 \leq i \leq m$. Then, for every real $\beta \geq C$ with $\beta b \leq c$, the following hold.*

(i) *For $1 \leq i \leq m$,*

$$\mathbb{P}(S^i_n \geq c\beta) \geq \frac{1}{4}e^{-C\beta^2} - \alpha \qquad \text{and} \qquad \mathbb{P}(S^i_n \leq -c\beta) \geq \frac{1}{4}e^{-C\beta^2} - \alpha\,.$$

(ii) *If $|\langle S^i, S^j\rangle_n| \leq \delta$ on Ω for all $i \neq j$, and $\beta^2\delta + \beta^3 b \leq 1$, then*

$$\mathbb{P}\Bigl(\max_{1\leq i\leq m} S^i_n < c\beta\Bigr) \leq C\bigl(m^{-1}e^{C\beta^2} + \beta^2\delta + \beta^3 b + e^{C\beta^2}\sqrt{\alpha}\bigr)\,.$$

Proof. We take $c \leq 1/8$ small and C large, depending only on c_0 and C_0. Stop all the martingales just before any bracket would exceed $2C_0$. Brackets are predictable, so this is a stopping time, and on Ω it does not stop before n; so we may assume that $\langle S^i\rangle_n \leq 2C_0$, and then $V_{ij} := \langle S^i, S^j\rangle_n$ satisfies $|V_{ij}| \leq 2C_0$. For a martingale S with respect to the common filtration $(\mathcal{G}_t)$, with $S_0 = 0$ and increments at most b_1, and for real s, let $\psi(s)$ be the sum over $t < n$ of the logarithms of the conditional moment generating functions $\mathbb{E}(e^{s(S_{t+1}-S_t)} \mid \mathcal{G}_t)$, and let $W(s) := \exp\{sS_n - \psi(s)\}$, so that $\mathbb{E}W(s) = 1$. If $\langle S\rangle_n \leq 8C_0$ and $|s|b_1 \leq 1/4$, a conditional Taylor expansion gives $\psi(s) = s^2\langle S\rangle_n/2 + O(|s|^3 b_1)$, and

the same at $2s$; so $W(s)^2 = W(2s)\exp\{\psi(2s) - 2\psi(s)\}$ gives $\mathbb{E}W(s)^2 \le e^{Cs^2}$. Apply these bounds to S^i and $S^i + S^j$, and write W_i, ψ_i and W_{ij}, ψ_{ij} for the corresponding values at $s = \beta$. Then

$$\psi_i = \frac{\beta^2}{2} V_{ii} + O(\beta^3 b) \qquad \text{and} \qquad \psi_{ij} - \psi_i - \psi_j = \beta^2 V_{ij} + O(\beta^3 b)\,.$$

(i) On Ω, $V_{ii} \ge c_0$ gives $\psi_i \ge c_0\beta^2/4$, so $W_i \ge 1/2$ implies $S_n^i \ge c_0\beta/4 - (\log 2)/\beta \ge c\beta$. By the Paley–Zygmund inequality, $\mathbb{P}(W_i \ge 1/2) \ge (\mathbb{E}W_i)^2/(4\mathbb{E}W_i^2) \ge e^{-C\beta^2}/4$. The same argument applies to $-S^i$.

(ii) For $i \ne j$, the exponent in $W_iW_j = W_{ij}\exp\{\psi_{ij} - \psi_i - \psi_j\}$ is at most $\beta^2\delta + C\beta^3 b$ on Ω and at most $C\beta^2$ everywhere, so the Cauchy–Schwarz inequality on Ω^c gives $\mathbb{E}(W_iW_j) \le 1 + C(\beta^2\delta + \beta^3 b) + e^{C\beta^2}\sqrt{\alpha}$. Hence the average $\overline{W}$ of $W_1, \dots, W_m$ has mean 1 and variance at most $C(m^{-1}e^{C\beta^2} + \beta^2\delta + \beta^3 b + e^{C\beta^2}\sqrt{\alpha})$. On Ω, $\overline{W} \ge 1/2$ implies $W_i \ge 1/2$ for some i, hence $\max_i S_n^i \ge c\beta$ as in (i), and Chebyshev's inequality proves (ii). □

9.1. **Lower bounds on the radii and their difference in the plane.** Up to additive constants, $R_{\text{in}}(n) \le R_{\text{mom}}(n) \le R_{\text{out}}(n)$, and by (8.1) the difference $R_{\text{mom}}(n) - r_n$ is, to first order, proportional to $\mathcal{Q}_n$. So Theorem 1.3 (i) needs deviations of $\mathcal{Q}_n$ of about $(\log n)^{1/2}$ standard deviations with probability at least n^{-p}. Lemma 9.1 (i) gives them from the coarse bounds: the upper bound on each local time forces a positive fraction of the steps to occur at distance comparable to r_n, which bounds the bracket of $\mathcal{Q}$ below. For part (iv) we use $Z_n = X_n + \kappa\sum_{x\in A_n} u_x$, which is a martingale because the step from X_j has conditional mean $-\kappa I_j u_{X_j}$. The vectors u_x cancel over the sites inside the inner disk, so $|Z_n - X_n|$ is at most $\kappa(R_{\text{out}}(n)^2 - R_{\text{in}}(n)^2) + Cr_n$ when both radii lie between $r_n/2$ and $2r_n$. Lemma 9.1 (i) for the first coordinate of Z, whose bracket lies between two constant multiples of n on every path, gives part (iv)(a); the martingale law of the iterated logarithm for the same coordinate gives part (iv)(b), and the martingale central limit theorem for Z gives part (iv)(c).

Proof of Theorem 1.3 (i) *and* (ii). Comparison with unit cells gives $\sum_{|x|<r}|x| = d\omega_d r^{d+1}/(d+1) + O(r^d)$. Since every lattice site in $B(0, R_{\text{in}}(n))$ belongs to A_n and $A_n \subset B(0, R_{\text{out}}(n) + 1)$, taking $(d+1)$st roots gives, for every path,

$$R_{\text{in}}(n) \le R_{\text{mom}}(n) + C \qquad \text{and} \qquad R_{\text{out}}(n) \ge R_{\text{mom}}(n) - C\,. \tag{9.2}$$

For $d = 2$, the two signs in (8.3) therefore give (1.4), with constant $\sqrt{2}/\sqrt{20\pi\kappa} = 1/\sqrt{10\pi\kappa}$.

For the polynomial lower bounds, let $r = r_n$. On Ω_n the stopped martingale $\widehat{\mathcal{Q}}$ of Section 5.1, whose increments are at most $4Kr$, agrees with $\mathcal{Q}$ through time n, and the coarse bounds force at least $n/2$ departures outside $B(0, cr)$; since $\kappa < 1/2$, (8.4) gives $c_0 r^5 \le \langle\widehat{\mathcal{Q}}\rangle_n \le C_0 r^5$ on Ω_n. Lemma 9.1 (i) for $\widehat{\mathcal{Q}}/r^{5/2}$, with $b = 4Kr^{-3/2}$, $\alpha = \mathbb{P}(\Omega_n^c)$ and $\beta = a_p\sqrt{\log n}$ for a small $a_p > 0$, shows that the events $\{\pm\mathcal{Q}_n \ge ca_p r^{5/2}\sqrt{\log n}\} \cap \Omega_n$ have probability at least n^{-p}. On Ω_n, $|X_n|^2 = O(r^2)$ and $R_{\text{mom}}(n) = O(r)$, so (8.1) and $R_{\text{mom}}(n)^3 - r^3 = (3/(2\pi))(\sum_{x\in A_n}|x| - n/(2\kappa))$ turn these events into $\pm(R_{\text{mom}}(n) - r) \ge c\sqrt{r\log n}$, and (9.2) proves (1.3). □

Proof of Theorem 1.3 (iv). Let $d = 2$, $V_n = \sum_{x\in A_n} u_x$, and $Z_n = X_n + \kappa V_n$. This martingale has increments bounded by 2, and the conditional variance computed for (8.4) gives, for $q \in \mathbb{R}^2$,

$$\langle q\cdot Z\rangle_n = \frac{|q|^2 n}{2} - \kappa^2 \sum_{x\in A_n} (q\cdot u_x)^2\,, \tag{9.3}$$

so $Y = e_1 \cdot Z$ satisfies $(1/2 - \kappa^2)n \le \langle Y\rangle_n \le n/2$ on every path. Suppose that $r_n/2 \le R_{\text{in}}(n) \le R_{\text{out}}(n) \le 2r_n$. The vectors u_x at the lattice sites with $|x| < R_{\text{in}}(n)$ sum to zero, and comparing the cells of the other sites of A_n with the annulus $\{R_{\text{in}}(n) - 1 \le |v| \le R_{\text{out}}(n) + 1\}$, on which $\int (e\cdot u_v)_+\,\mathrm{d}v = R_{\text{out}}(n)^2 - R_{\text{in}}(n)^2 + O(r_n)$ for every unit vector e, gives

$$|V_n| \le R_{\text{out}}(n)^2 - R_{\text{in}}(n)^2 + Cr_n\,. \tag{9.4}$$

Lemma 9.1 (i) for $Y/\sqrt{n}$, with $\alpha = 0$ and $\beta = a_p\sqrt{\log n}$ for a small $a_p > 0$, shows that $\{Y_n \geq ca_p\sqrt{n\log n}\} \cap \Omega_n$ has probability at least n^{-p}. On this event, for large n, both radii lie in $[r_n/2, 2r_n]$, and $|X_n| = O(r_n)$, (9.4) and $n = (4\pi\kappa/3)r_n^3$ give $R_{\text{out}}(n) - R_{\text{in}}(n) \geq c\sqrt{n\log n}/r_n - C \geq c\sqrt{r_n\log n}$, which proves (1.6).

By Theorem 1.1, $|A_n| = o(n)$, $X_n = o(\sqrt{n})$ and $R_{\text{in}}(n) + R_{\text{out}}(n) \sim 2r_n$ almost surely. So $\langle q \cdot Z\rangle_n/n \to |q|^2/2$ by (9.3), and the limit theorems stated at the beginning of Section 8, with $s_n = \sqrt{n}$, give $\limsup Y_n/\sqrt{n\log\log n} = 1$ almost surely and $Z_n/\sqrt{n} \to N \sim \mathcal{N}(0, \frac{1}{2}I_2)$ in distribution. By (9.4), almost surely

$$\frac{|Z_n|}{\sqrt{n}} \leq \sqrt{\frac{3\kappa}{\pi}}\,\frac{R_{\text{out}}(n) - R_{\text{in}}(n)}{\sqrt{r_n}}(1 + o(1)) + o(1)\,.$$

The first limit therefore gives $\limsup(R_{\text{out}}(n) - R_{\text{in}}(n))/\sqrt{r_n\log\log n} \geq \sqrt{\pi/(3\kappa)}$, and, since $\mathbb{P}(|N| \leq t) = 1 - e^{-t^2}$, the second shows that $\limsup \mathbb{P}(R_{\text{out}}(n) - R_{\text{in}}(n) \leq a\sqrt{r_n}) \leq 1 - e^{-3\kappa a^2/\pi}$. □

9.2. **Upper bounds on the local times away from the boundary.** At sites y with $|y| \leq \theta r_n$, where $0 < \theta < 1$, the set $D_n \setminus B(0, R_{\text{in}}(n))$ has small potential, because it has small volume and lies at distance of order r_n from y; only the error in (5.7) remains. In the plane the resulting bound improves Theorem 1.2 (iii).

Proposition 9.2. *Let Ψ be the Euclidean norm, let $0 < \kappa < 1/d$, and let $0 < \theta < 1$ and $p > 0$. There exists $C(d, \kappa, \theta, p) < \infty$ such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$,*

$$\max_{\substack{y\in\mathbb{Z}^d\\ |y|\leq\theta r_n}} \bigl|\ell_n(y) - 2d\kappa(r_n - |y|)\bigr| \leq C\begin{cases}\sqrt{r_n}\log n, & d = 2,\\ \sqrt{r_n\log n}, & d \geq 3\end{cases}. \tag{9.5}$$

The same bound holds for all sufficiently large n almost surely.

Proof. Intersect the events of Theorem 1.2 and of Section 5.1, with exponent $\max\{p, 2\}$ in both, and let $E = D_n \setminus B(0, R_{\text{in}}(n))$ as in Section 6. For large n, every y with $|y| \leq \theta r_n$ lies at distance at least $c_\theta r_n$ from E, and $|E| \leq Cr_n^d q_n$, so $|U_E(y)| \leq C_\theta |E| r_n^{1-d} \leq C_\theta r_n q_n$. With $|R_{\text{in}}(n) - r_n| \leq Cr_n q_n$ and (1.10), this gives, uniformly over $|y| \leq \theta r_n$,

$$\bigl|U_{D_n}(y) - 2d\kappa(r_n - |y|)\bigr| \leq C_\theta r_n q_n\,, \tag{9.6}$$

and $r_n q_n$ is at most the right side of (5.7), which proves (9.5). The Borel–Cantelli lemma gives the almost-sure statement. □

9.3. **Lower bounds on the local times near the origin.** By (3.9) and (9.6), on Ω_n the deviation $\ell_n(y) - 2d\kappa(r_n - |y|)$ at a site y with $|y| \leq r_n/2$ equals $-\mathcal{M}_n^y$ up to $O(r_n q_n + \log n)$. So it suffices to find, among about $r_n^{1/8}$ sites at mutual distances at least $r_n^{1/4}$, one at which $-\mathcal{M}_n^y$ exceeds a multiple of $(\log r_n)^{1/2}$ standard deviations. In the plane the martingales at nearby sites are strongly correlated (see the proof of Theorem 8.3), so we use differences of martingales at two sites at distance about $r_n^{1/8}$. The normalized martingales then have brackets bounded above and below and cross brackets of order at most $r_n^{-1/4}$ (Lemma 9.3), and Lemma 9.1 (ii) applies.

Recall the martingales $\mathcal{M}^f$ of Section 3.1 and the function $\Gamma(f_1, f_2)$ of (8.7). Let $\sigma_n := (r_n\log r_n)^{1/2}$ and $k := \lfloor r_n^{1/8}\rfloor$ when $d = 2$, and $\sigma_n := r_n^{1/2}$ when $d \geq 3$. Let $m := \lfloor r_n^{1/8}\rfloor$, and for $1 \leq i \leq m$ let

$$y_i := i\lceil r_n^{1/4}\rceil e_1\,, \qquad f_i := \begin{cases} g_{y_i+ke_1} - g_{y_i}, & d = 2,\\ g_{y_i}, & d \geq 3\end{cases} \qquad \text{and} \qquad S^i := -\frac{\mathcal{M}^{f_i}}{\sigma_n}\,.$$

Then $|y_i| \leq m\lceil r_n^{1/4}\rceil \leq r_n^{1/2}/2$ for large n.

Lemma 9.3. *Let Ψ be the Euclidean norm, and let $0 < \kappa < 1/d$. There exist $0 < c_0(d,\kappa) \le C_0(d,\kappa)$ and $C(d,\kappa) < \infty$ such that, for all sufficiently large n, on Ω_n, for $1 \le i \le m$ and $1 \le j \le m$ with $j \ne i$,*

$$c_0 \le \langle S^i \rangle_n \le C_0 \qquad \text{and} \qquad |\langle S^i, S^j \rangle_n| \le C r_n^{-1/4} .$$

Proof. Write $r = r_n$, and let $L := \log r$ if $d = 2$ and $L := 1$ if $d \ge 3$, so that $\sigma_n^2 = rL$. By (3.2) and (3.3), the increments of f_i at x are $O((1+|x-y_i|)^{1-d})$ if $d \ge 3$, and in the plane $O((1+|x-y_i|)^{-1} + (1+|x-y_i-ke_1|)^{-1})$ for $|x-y_i| \le 3k$ and $O(k|x-y_i|^{-2})$ otherwise. Since $2|ab| \le a^2 + b^2$ and Δf_i is supported at the poles, which lie in $B(0,\sqrt{r})$,

$$\sum_x |\Gamma(f_i, f_j)(x)| \le CL \qquad \text{and} \qquad \sum_{|x|>2\sqrt{r}} |\Gamma(f_i, f_j)(x)| \le \frac{CL}{\sqrt{r}} ,$$

and the first-departure corrections to the bracket total $O(L)$. Let $\Sigma_{ij} := (2d\kappa/L) \sum_x \Gamma(f_i, f_j)(x)$. On Ω_n, (9.1) gives $\ell_n(x) = 2d\kappa r + O(\sqrt{r}(\log n)^{3/2})$ for $|x| \le 2\sqrt{r}$, and $\ell_n \le Cr$ everywhere, so, uniformly in i, j,

$$\big|\langle S^i, S^j \rangle_n - \Sigma_{ij}\big| \le \frac{C}{rL}\big(L + \sqrt{r}(\log n)^{3/2} L + \sqrt{r} L\big) \le C r^{-1/2} (\log n)^{3/2} = o(r^{-1/4}) .$$

For $d \ge 3$, as in the proof of Theorem 8.3, $\Sigma_{ij} = 2d\kappa(2G(y_i - y_j) - \mathbf{1}_{\{i=j\}})$, whose diagonal is $2d\kappa(2G(0) - 1) > 0$ and whose off-diagonal entries are $O(r^{(2-d)/4})$. In the plane, (8.11) applies to f_i, f_j, whose values and increments at distance R are $O(k/R)$ and $O(k/R^2)$; since Δf_i and Δf_j have disjoint supports for $i \ne j$, it gives, with $z = y_i - y_j$,

$$\sum_x \Gamma(f_i, f_j)(x) = 2\big(g(z + ke_1) + g(z - ke_1) - 2g(z)\big) - 2\mathbf{1}_{\{i=j\}} .$$

So $\Sigma_{ii} = 4\kappa(4g(ke_1) - 2)/\log r \to 4\kappa/\pi$, and, for $i \ne j$, $\Sigma_{ij} = O(k^2/|z|^2) = O(r^{-1/4})$ by (3.2). □

Proof of Theorem 1.3(iii). By (3.3), the increments of S^i are at most $b = Cr_n^{-1/2}$. Lemma 9.1 (ii) with $\Omega = \Omega_n$, the brackets of Lemma 9.3, $\delta = Cr_n^{-1/4}$, and $\beta = a\sqrt{\log r_n}$ for a small $a > 0$, so that $m^{-1} e^{C\beta^2} \le r_n^{-1/16}$, gives

$$\mathbb{P}\left(\max_i S_n^i < ca\sqrt{\log r_n}\right) \le C r_n^{-1/16} .$$

On Ω_n, by (3.9) and (9.6) with $\theta = 1/2$, the deviation $\ell_n(y) - 2d\kappa(r_n - |y|)$ equals $-\mathcal{M}_n^y + o(\sigma_n\sqrt{\log r_n})$, uniformly over $|y| \le r_n/2$. So, if $S_n^i \ge ca\sqrt{\log r_n}$, then for $d \ge 3$ the deviation at y_i is at least $c\sqrt{r_n \log n}$, and for $d = 2$ the deviations at $y_i + ke_1$ and y_i differ by at least $c\sqrt{r_n}\log n$, so one of them is at least $(c/2)\sqrt{r_n}\log n$ in absolute value. All these sites lie in $B(0,\sqrt{r_n})$, and the failure probability is at most $Cr_n^{-1/16} + O(n^{-10}) \le n^{-c}$, which proves (1.5). □

9.4. **Logarithmic lower bounds for the radii.** If D_n contains $B(0, r_n - h)$, then the walk has visited every site of an annulus of width of order h inside the sphere of radius r_n. If $R_{\text{out}}(n) \le r_n + h$, then the martingale $\mathcal{Q}$ of (8.1) shows that, except with probability at most $\exp\{-cr_n^{d-3}h^2\}$, the walk has visited at least $cr_n^{d-1}h$ sites of an annulus of width of order h near this sphere; this probability is small when $d \ge 3$ and h has order $\log n$. From each such site the walk leaves $B(0, r_n + h)$ along a straight path with conditional probability at least e^{-Ch}, so the probability that it follows none of these paths from cr_n^{d-1} first visits on disjoint time intervals is at most $\exp\{-cr_n^{d-1}e^{-Ch}\}$.

Proposition 9.4 (Lower bounds for the radii)**.** *Let Ψ be the Euclidean norm, and let $0 < \kappa < 1/d$. There exist $c(d,\kappa) > 0$, $C(d,\kappa) < \infty$ and $h_0(d,\kappa) > 0$ such that the following hold.*

(i) Tail bounds: *for all sufficiently large n and every real h with $h_0 \le h \le r_n/8$,*

$$\mathbb{P}\big(R_{\text{in}}(n) \ge r_n - h \text{ and } R_{\text{out}}(n) \le r_n + h\big) \le \exp\big\{-cr_n^{d-1}e^{-Ch}\big\}, \tag{9.7}$$

$$\mathbb{P}\big(R_{\text{out}}(n) \le r_n + h\big) \le \exp\big\{-cr_n^{d-1}e^{-Ch}\big\} + \exp\big\{-cr_n^{d-3}h^2\big\}. \tag{9.8}$$

(ii) Almost-sure bounds: *almost surely, for all sufficiently large n,*

$$\max\{r_n - R_{\text{in}}(n), R_{\text{out}}(n) - r_n\} > c\log n, \tag{9.9}$$

and, if $d \ge 3$, also $R_{\text{out}}(n) - r_n > c\log n$.

In the plane the last term in (9.8) is not small when h has order $\log n$.

Proof. Write $r = r_n$ and $\Lambda = \{x \in \mathbb{Z}^d : r - 2h < |x| \le r + h\}$. Every transition probability is at least $1/(2d) - \kappa/2 > 0$, so from each site of Λ the walk leaves $B(0, r+h)$ in $T = \lceil 4h\sqrt{d}\rceil$ outward steps along a coordinate of largest absolute value with conditional probability at least e^{-Ch}. Run such trials from successive first visits to Λ before time $n - T$, on disjoint time intervals of length T. If $|A_n \cap \Lambda| \ge c_1 r^{d-1}h$, then at least cr^{d-1} trials start, because each interval contains at most $T + 1$ first visits; so, for every fixed $c_1 > 0$,

$$\mathbb{P}\big(R_{\text{out}}(n) \le r + h \text{ and } |A_n \cap \Lambda| \ge c_1 r^{d-1}h\big) \le \exp\{-cr^{d-1}e^{-Ch}\}. \tag{9.10}$$

For c_1 small and h_0 large, the sites with $r - 2h < |x| < r - h - 1$, which are in A_n when $R_{\text{in}}(n) \ge r - h$, number at least $c_1 r^{d-1}h$; this proves (9.7). If instead $R_{\text{out}}(n) \le r + h$ and $|A_n \cap \Lambda| < c_1 r^{d-1}h$, then $|x| \le \int_{C_x} |v|\,\mathrm{d}v$ and the inclusion of the cells with $|x| \le r - 2h$ in $B(0, r-h)$ give $\sum_{x\in A_n} |x| \le n/(2\kappa) - c_2 r^d h + 2c_1 r^d h$. For $c_1 \le c_2/4$, (8.1) gives $\mathcal{Q}_n \le -cr^d h$, and Azuma's inequality for $\mathcal{Q}$ stopped on leaving $B(0, r+h)$, with increments at most $5r$, bounds the probability of this event by $\exp\{-cr^{2d}h^2/(nr^2)\} = \exp\{-cr^{d-3}h^2\}$. With (9.10), this proves (9.8). Taking $h = c\log n$ with c small makes the first tail summable in every dimension and the second for $d \ge 3$, and the Borel–Cantelli lemma proves (ii). □

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